Multi-Task Semi-Supervised Learning with Automatically Generated Auxiliary Labels for Dental CBCT Segmentation

A Thesis

Presented to the

School of Information Technology

Mapua University

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Computer Science

By

Alisandro N. Domingo

Jherrie Nathaniel C. Millena

Ryan Osmon S. Patenia

March 2026

RECOMMENDATION SHEET

This is to certify that I have supervised the preparation of and read the thesis proposal prepared by **Alisandro N. Domingo, Jherrie Nathaniel C. Millena, and Ryan Osmon S. Patenia** entitled **Multi-Task Semi-Supervised Learning with Automatically Generated Auxiliary Labels for Dental CBCT Segmentation** and that the said paper has been submitted for final examination by the Oral Examination Committee.

_____

Dr. Lysa V. Comia
Thesis Adviser

Multi-Task Semi-Supervised Learning with Automatically Generated Auxiliary Labels for Dental CBCT Segmentation i

RECOMMENDATION SHEET ii

LIST OF FIGURES vii

LIST OF TABLES viii

Chapter 1 1

1.1. Context of the Study 3

1.2. The Problem, Gap or Opportunity 8

1.3. Research Questions 10

1.4. Research Objectives 11

1.5. Significance of the Study 12

1.6. Scope and Limitations (or Delimitations) 13

Chapter 2 17

2.1. Clinical Imperatives in AI-Driven Dental Imaging 17

2.1.1 Epidemiology and Clinical Burden of Dental Imaging 17

2.1.2 Anatomical Complexity of Maxillofacial Structures 19

2.1.3 The Diagnostic Superiority of CBCT Over Two-Dimensional Radiography 21

2.2. Deep Learning Architectures for Medical Image Segmentation 23

2.2.1 Convolutional Neural Networks: The Established Foundation 23

2.2.2 Transformers: Global Attention at Computational Cost 25

2.2.3 State Space Models: Efficient Long-Range Modeling 27

2.3. Multi-Task Learning in Medical Image Analysis 29

2.3.1. Evidence for Multi-Task Benefits in Dental Segmentation 30

2.3.3. The Critical Limitation: Annotation Multiplication 32

2.3.4 Automatic Auxiliary Label Generation: Methodological Foundation 33

2.4. Semi-Supervised Learning for Medical Image Segmentation 34

2.4.1. Theoretical Foundations and the Consistency Assumption 34

2.4.2. Mean Teacher Framework 35

2.4.3. Semi-Supervised Dental CBCT Segmentation: Benchmarks and State of the Art 37

2.5. Evaluation Metrics for Dental CBCT Segmentation 38

2.6. Synthesis and Research Gaps 40

Chapter 3 43

3.1 Conceptual Framework 43

3.2. Research Design 44

3.3. Dataset and Data Preparation 46

3.3.1 ToothFairy3 Dataset 46

3.3.2 STS-3D-Tooth Dataset 47

3.3.3 Dataset Splitting 48

3.3.4 ToothFairy3 Anatomical Region Classification and Clinical Significance .. 49

3.3.5 Preprocessing Pipeline 51

3.4. Automatic Auxiliary Label Generation 52

3.4.1 Tooth Apex Landmark Extraction via Morphological Skeletonization..... 53

3.4.2 Nerve Centerline Extraction via Distance Transform and Thinning 55

3.5. Model Architecture: Multi-Task U-Mamba2..... 56

3.5.1 Base U-Mamba2 Architecture 56

3.5.2 Multi-Task Extension: Three Task-Specific Decoder Heads 58

3.6. Semi-Supervised Learning Framework: Mean Teacher 59

3.6.1 Mean Teacher Training Protocol and Multi-Task Consistency Regularization 59

3.6.2 Task-Specific Loss Functions and Multi-Task Weighting 61

3.6.3 Data Augmentation Strategy: Weak Versus Strong Perturbations 62

3.7. Training Protocol and Hyperparameters 63

3.8. Evaluation Metrics and Analysis 64

3.8.1. Primary Metrics 64

3.8.2. Structure Specific Analysis..... 66

3.9. Role of Domain Experts..... 67

3.10. Deployment Scope 68

3.11. Timeline and Feasibility 70

3.12. Ethical Considerations 71

APPENDICES 75

REFERENCES 77

LIST OF FIGURES

Figure 1. Anatomical Complexity of the Maxillofacial Region. Reprinted from [40]. ... 20

Figure 2. 2D Radiography vs CBCT Scan Diagnostic Comparison. Reprinted from [41] ... 22

Figure 3. U-Net Encoder-decoder Architecture with Skip Connections. Retreived from [43] ... 24

Figure 4. U-Mamba2 model Architecture. Retrieved from [16] ... 28

Figure 5. Multi-Task Framework ... 30

Figure 6. Mean Teacher Pipeline for Medical Image Analysis ... 36

Figure 7. Conceptual Framework Diagram ... 44

Figure 8. CBCT Scan in different views ... 47

Figure 9. Automatic Auxiliary Label Generation Pipeline ... 53

Figure 10. Proposed multi-task U-Mamba2 architecture extending the base U-Mamba2 ... 58

Figure 11. Mean Teacher Pipeline ... 60

LIST OF TABLES

Table 1. Architecture Comparison on Dental CBCT Segmentation 27

Table 2. Experimental Configuration Summary 45

Table 3. Dataset Split Summary 48

Table 4. ToothFairy3 Anatomical Region Classification 49

Table 5. Evaluation Size Categories for Disaggregated Metric Reporting 51

Table 6. Auxiliary Task Source Classes 54

Table 7. Automatic Auxiliary Label Generation Performance Summary 55

Table 8. Data Augmentation Strategy for Weak and Strong Perturbations 62

Table 9. Complete Hyperparameter Specification 63

Table 10. Anatomical Size Categories and Clinical Significance 66

Table 11. Evaluation Metrics and Clinical Acceptance Thresholds 66

Table 12. Complete 15-Week Study Timeline 70

Chapter 1

INTRODUCTION

Cone-Beam Computed Tomography (CBCT) is the primary three-dimensional imaging modality in modern dental practice. Unlike conventional two-dimensional radiographs, CBCT produces volumetric scans that allow clinicians to visualize teeth, jawbones, neurovascular structures, and surrounding anatomies with sub-millimetre resolution [1], [2]. This level of detail is essential for procedures such as implant placement, orthodontic planning, and endodontic diagnosis. However, making clinical use of CBCT data requires segmentation: the process of identifying and delineating specific anatomical structures within the three-dimensional volume. Comprehensive segmentation of a single CBCT scan involves labelling over 70 distinct structures and typically requires 25–45 minutes of expert effort per scan when performed manually [3], [4]. For small neurovascular structures such as the inferior alveolar canal, inter-observer agreement measured by the Dice Similarity Coefficient (DSC) can be as low as 72–78%, reflecting the difficulty of consistent manual delineation [4].

Recent advances in deep learning have demonstrated that automated segmentation can approach expert-level accuracy while reducing analysis time to under one minute per scan [5], [6]. Among these, the U-Mamba2 architecture—which integrates Mamba2 state space model blocks within a U-Net encoder-decoder backbone—secured a top-three position in the ToothFairy3 multi-structure CBCT segmentation challenge at MICCAI 2025 [7]. Its semi-supervised extension, U-Mamba2-SSL, achieved 91.7% DSC for tooth and pulp segmentation using only 8% labelled data in the STSR 2025 challenge, demonstrating that strong performance is possible with minimal annotation [8].

A real gap remains between benchmark numbers and day-to-day clinical use. The core problem is annotation: training good segmentation models requires large volumes of expertly labeled scans, and producing dense voxel-wise labels is expensive enough to be practically prohibitive. The STS-Tooth dataset illustrates the scale of the problem — it holds 148,400 CBCT scans, but only 8,800 (5.9%) are fully annotated, and generating those annotations took 20 trained dental practitioners plus 6 experienced dentists an entire year [9].

Semi-supervised learning (SSL) attacks this bottleneck by making use of large pools of unlabeled data alongside small labeled sets [10], [11]. Multi-task learning (MTL) offers a complementary benefit: training a model to predict auxiliary anatomical targets landmarks, centerlines, boundaries, alongside the primary segmentation tends to improve the representations it learns [12], [13], [14]. The catch is that existing MTL approaches require separate manual labels for every auxiliary task, which can double or triple per-scan annotation time [13], [14], [15]. The method designed to reduce labeling burden ends up adding to it.

This thesis examines a different approach, one not previously explored: automatically generating auxiliary task labels from existing segmentation masks using morphological image processing. Tooth apex landmarks can be derived via three-dimensional skeletonization; nerve centerlines via distance-transform-based thinning. Both operations take seconds per scan and require no additional manual annotation. The proposed framework combines this label-generation approach with Mean Teacher SSL and the U-Mamba2 architecture, aiming for annotation-efficient dental CBCT segmentation at

no incremental labeling cost. This chapter lays out the rationale, states the research questions and objectives, and defines the scope and limitations of the study.

1.1. Context of the Study

CBCT has changed how dentists use images when diagnosing problems with the jaw (maxillofacial region) because of the high-resolution 3D images that CBCT provides (isotropic voxels of ~0.08-0.4 mm^3) [1], [2] and assist more accurately than traditional 2D X-ray images (in one study of 320 patients, CBCT identified 88.2% of periapical lesions vs. 64.7% with intra-oral films ($p < 0.001$)) especially in the posterior areas where superimposing on projection images masked pathology [16]. In implant placement studies, retrospective analysis of 120 cases revealed that CBCT imaging identified 34% more periapical lesions than periapical films due to eliminating superimposition thereby allowing for identification of bone that would have been less than 1 millimetre in thickness as well as identifying buccal-lingual cortex breaches [17]. For AI-assisted CBCT studies, 2023 meta-analysis pooled segmented results, which gives a mean overall Dice score of 0.92 (95% CI: 0.88-0.95). Performance on segmentation varied widely by structure, e.g., 0.55 (very small) for neurovascular structures, and 0.98 (large) for airway structures [18].

The availability of more sophisticated data has not translated into more clinician capability to correctly interpret CBCT images since they contain hundreds (200 to 600) of axial slices with varying thicknesses. In order to evaluate a scan accurately, a physician must evaluate each slice in three planes, and the physician performing the segmentation must take an average of 25-45 minutes per scan to perform the segmentation necessary for treatment planning. Artificial intelligence tools are able to reduce the amount of time spent doing image processing significantly. In fact, validation studies have shown that by using

artificial intelligence for automatic segmentation of objects within the scanned volume of a chest CT, it is possible to reduce the amount of time taken for a radiologist to do their analysis from approximately 25 minutes down to less than one minute while still producing results that are in agreement within 92-94%, when compared to the performance of expert radiologists, in all five anatomical regions being evaluated. Each clinic that provides 50-100 CBCT scans per week will be able to increase the number of hours available to a radiologist to perform complex diagnostic reasoning and communicate with patients and their families about the radiology findings.

Automating segmentation of the maxillofacial region is technically demanding because of the number and diversity of structures involved. ToothFairy3 annotates 77 distinct anatomical classes: 32 individual permanent teeth (FDI 11–48), 32 corresponding tooth pulp cavities, 2 jawbones (mandible, maxilla), 2 inferior alveolar canals, 2 maxillary sinuses, 2 mandibular incisive canals, 1 lingual canal, 1 pharynx, and 3 dental restoration classes (bridge, crown, implant) [3], [43]. The class imbalance across these structures is extreme: jawbones may occupy 30–40% of a scan's volume while teeth account for 15–20%, pulp cavities collectively constitute less than 1%, and the five neurovascular canal structures—the inferior alveolar canals, incisive canals, and lingual canal—together occupy less than 0.1% of the total volume [3], [7]. Models trained without accounting for this imbalance tend to perform well on large structures while failing on the small ones that matter most for surgical safety.

The anatomical complexity of neurovascular structures compounds this challenge. The inferior alveolar canal occasionally deviates more than two millimetres from its expected path, may bifurcate at multiple points, and often presents with indistinct margins

in areas of low-density bone [17]. Mental foramina and incisive canals can vary by up to three millimetres from their anticipated location relative to standard anatomical landmarks [17]. These variations directly affect surgical risk: inaccurate nerve localisation during implant placement can cause permanent sensory loss. Reported Dice scores for inferior alveolar nerve segmentation span a wide range, from approximately 55% for baseline U-Net models to 85–90% for more advanced architectures incorporating active learning or multi-label training strategies [19], [20]. Patient variability adds further difficulty, including mixed dentition in paediatric cases with up to 52 teeth at various developmental stages, dental pathology such as caries and periapical disease that alters expected morphology, and beam-hardening artefacts from metallic restorations that can obscure local anatomy even for experienced clinicians [16], [21].

A central barrier to progress is the cost of producing high-quality annotations. The STS-Tooth dataset illustrates the scale of this problem: annotating 8,800 CBCT scans required a team of 20 dental practitioners working under the supervision of 6 experienced dentists for approximately one year, and this still represented only 5.9% of the 148,400 available scans [9], [44]. Annotating at the 77-class granularity of ToothFairy3—where each individual tooth, its pulp cavity, and every neurovascular canal must be separately delineated—further increases per-scan effort compared to earlier datasets with fewer classes [3], [43]. This annotation scarcity concentrates AI development capability within well-resourced institutions that can afford large annotation teams, while smaller practices and resource-limited settings—where AI assistance could have the greatest clinical impact—remain excluded. Beyond diagnosis, the main value of automated segmentation lies in producing three-dimensional models for guided surgery, orthodontic simulation, and

computer-aided planning; manual 3D modelling currently takes 25–45 minutes per scan for experts, versus under one minute with state-of-the-art automated methods, enabling same-visit digital workflows that would otherwise be impractical [5].

Volumetric medical imaging segmentation architectures have transitioned from convolutional neural networks (CNNs) and attention based transformers to state space models (SSMs). Examples of strong baselines are CNNs, including nnU-Net and the v2 version of nnU-Net where nnU-Net achieved 86.1% mean Dice scores on ToothFairy3. Additionally, many of the newer CNNs can produce comparable or superior results when adequately scaled and tested for standardized training (see [23], [24] for additional examples). While transformers methodically provide a global context using a self-attention mechanism, their performance begins to diminish with increasing resolution of 3D data due to the quadratic relationship between the amount of processing needed to maintain global context (i.e., self-attention cost) and the length of the sequence (see [26], [27] for additional details). In contrast, state space methods (e.g., Mamba and Mamba2) resolve the problem of providing a global context with a linear cost. The performance of the U-Mamba2 system, which is comprised of multiple Mamba2 blocks included in a 7-stage encoder-decoder architecture, resulted in an 87.3% mean Dice score on the ToothFairy3 image with an inference time of 6.81 seconds on an NVIDIA T4 system. Therefore, U-Mamba2 represents an optimal trade-off of accuracy and efficiency for use in clinical workstations [7], [28], [32]. Hybrid structure designs, which combine the ability of CNNs to detect local features at a high resolution with the ability of Mamba2 to model global features at the decreasing data locations (i.e., the bottleneck), are well suited for CBCT. It

creates global relationships between structures that are separated spatially (i.e. the tooth apex is located 20-30mm away from the inferior alveolar canal) and can aid with diagnosis.

To address annotation scarcity, semi-supervised learning (SSL) leverages unlabelled scans alongside a limited set of labelled examples. Mean Teacher frameworks, where a student network learns from gradient updates and a teacher network is the student's exponential moving average, enforce prediction consistency between strongly and weakly augmented views and form a practical SSL backbone [33]. Recent improvements add uncertainty-aware pseudo-labelling, cross-teacher ensembles, and transformer backbones to this paradigm [34], [35]. The MICCAI STS series has driven rapid progress: STS 2023 showed that combining ~30 labelled scans with 300 unlabelled ones can yield clinically acceptable performance through pseudo-labelling and consistency regularisation [36], and subsequent analyses highlighted the advantages of entropy-guided Mean Teacher models and multi-stage self-training [37]. STS 2024 extended evaluation to many tooth instances and deciduous dentitions [38], and U-Mamba2-SSL recently reported 91.7% Dice on the STSR 2025 tooth and pulp challenge using only 8% labelled data via a three-stage strategy incorporating disruptive autoencoder pre-training, consistency regularisation, and pseudo-labelling [8].

Multi-task learning (MTL) offers another route to better generalisation by training models to predict related outputs jointly—forcing shared encoders to learn richer features and acting as an implicit regulariser [12], [39], [40], [41]. In dental CBCT, MTL has shown consistent gains: joint tooth region and surface prediction produced a 3.8% Dice improvement [13], hierarchical semantic–instance–canal approaches gave 4.2% gains [14], and pairing canal segmentation with boundary refinement yielded 5.3% improvement [15].

The trade-off is that each auxiliary task often needs its own manual annotation, increasing per-scan labelling time substantially—Chen et al.'s dual-task method raised annotation time by 33–50%, and other multi-task frameworks have multiplied annotation effort—creating a paradox where methods that improve performance also increase the annotation burden they aim to eliminate.

Taken together, these trends define the practical and technical landscape for automated CBCT interpretation: richer imaging data and advanced model architectures can greatly accelerate and improve clinical workflows, but progress is constrained by annotation costs, difficult small-structure segmentation, and patient variability. Semi-supervised and multi-task strategies reduce label dependence and increase robustness, yet their deployment must carefully balance the gains in performance against annotation overhead so that AI tools deliver broad clinical benefit rather than concentrating capability in a few resource-rich centres.

1.2. The Problem, Gap or Opportunity

Current multi-task learning approaches for dental CBCT segmentation require two to three times the annotation budget of single-task models [13], [14], [15]. In practice, this means sourcing landmark, boundary, and structural annotations on top of the primary segmentation masks—expensive enough that MTL is mostly applied where labelled data is already plentiful. The settings where it would most help, those with scarce labels and limited annotation resources, are precisely the ones it cannot reach. Whether auxiliary labels generated automatically from existing segmentation masks—via morphological operations that take seconds per scan—could replicate those gains has not been tested in the dental imaging literature.

SSL and MTL for dental CBCT have also developed mostly in parallel rather than in combination. SSL studies in this domain [8], [31], [32], [33] focus on single-task segmentation and do not incorporate anatomical priors from auxiliary tasks. MTL studies [13], [14], [15] operate in fully supervised settings and do not leverage unlabelled data. Combining the two approaches raises real optimization challenges: consistency regularization must hold across multiple task outputs simultaneously, auxiliary pseudo-labels carry noise characteristics different from primary segmentation pseudo-labels, and task weighting becomes harder to calibrate when labelled data is limited. The interactions between SSL and MTL for dental CBCT segmentation remain largely unexplored.

There is also a gap in evaluation methodology, particularly for nerve canal segmentation. Volumetric Dice and the 95th-percentile Hausdorff Distance (HD95) are the standard metrics for medical image segmentation, but they do not capture whether tubular structures remain topologically intact. A predicted nerve canal that is split into disconnected fragments can still achieve a reasonable DSC while being clinically useless for implant planning—a dentist cannot safely plan around a nerve that appears fragmented in the model output. The centerline Dice (clDice) metric, introduced by Shit et al. [36], evaluates overlap between the morphological skeletons (centerlines) of the predicted and ground-truth structures, directly measuring topological connectivity. clDice is standard in vascular and airway segmentation but has not been applied to dental CBCT evaluation. Additionally, many MTL studies report only aggregate metrics across all structures, making it difficult to determine which auxiliary tasks are actually contributing useful learning signals and for which anatomical categories.

Morphological operations offer a practical path forward. Tooth apex landmarks can be extracted through three-dimensional skeletonization of tooth masks, where the algorithm iteratively removes surface voxels while preserving topology until only a one-voxel-wide skeleton remains, with endpoints representing root apices. Nerve centerlines can be extracted via distance-transform-based thinning, where the distance transform assigns each interior voxel its distance to the nearest boundary, and iterative thinning then traces the medial axis through the canal structure [36], [37]. Both operations produce clinically meaningful primitives from annotations that must already exist for the primary segmentation task—no extra labelling is required. If these automatically derived labels can recover even part of the performance gains associated with hand-annotated MTL tasks, multi-task learning becomes considerably more practical in resource-constrained clinical environments.

1.3. Research Questions

This study addresses the following research questions:

(1) How can a multi-task semi-supervised learning framework be designed to automatically generate auxiliary task labels (tooth apex landmarks and nerve centerlines) from existing segmentation masks at zero incremental annotation cost?

(2) Can the proposed framework be implemented and trained on 20% labeled data to produce measurable performance differences attributable to auxiliary task supervision?

(3) Does the proposed framework achieve clinically acceptable segmentation performance (DSC $\geq$ 0.80, HD95 $<$ 15 mm, clDice $\geq$ 0.75 for tubular

structures), and how do results vary across anatomical size categories (large, medium, small) and a Philippine external validation cohort?

1.4. Research Objectives

To address the annotation bottleneck in clinical AI deployment for dental imaging, this research aims to evaluate the effectiveness of multi-task semi-supervised learning for dental CBCT segmentation under annotation-scarce conditions. Specifically, this study seeks to:

(1) Design a multi-task semi-supervised learning framework integrating U-Mamba2 with Mean Teacher consistency regularization, wherein tooth apex landmarks and inferior alveolar nerve centerlines are automatically generated from segmentation masks as auxiliary task labels at zero incremental annotation cost.

(2) Implement the proposed framework on the ToothFairy3 benchmark using 20% of available scans as labelled data, training both a multi-task SSL configuration and a single-task SSL baseline for controlled comparison.

(3) Evaluate the performance of both configurations using:

   a. Dice Similarity Coefficient (DSC $\geq$ 0.80) on the test set, disaggregated across large, medium, and small anatomical categories,

   b. Hausdorff Distance 95th percentile (HD95 < 15 mm) on the same held-out test set, providing boundary localization accuracy complementary to volumetric overlap

c. Centerline Dice (clDice ≥ 0.75) for all tubular neurovascular structures, providing topology-preserving evaluation that captures structural connectivity [36].

1.5. Significance of the Study

The central obstacle to deploying AI segmentation tools in dental clinics is not algorithmic capability but annotation cost. Fully supervised CBCT segmentation pipelines require hundreds of expertly labelled scans [3], [9]. SSL reduces the labelled data requirement to approximately 20–30% of available data [8], [31], but existing MTL methods offset these efficiency gains by requiring additional auxiliary annotations [13], [14], [15]. The framework proposed in this thesis avoids that trade-off entirely: auxiliary labels are generated automatically from masks already required for the primary task, through morphological operations that run in seconds per scan. At this cost, capabilities that were previously impractical become feasible. A single dental practice could build a locally adapted segmentation model from 30–50 annotated cases. A community health centre with a limited budget could adopt CBCT-based AI without commissioning a large labelling effort.

The label generation pipeline is not specific to dentistry. The morphological operations used to extract tooth apex landmarks and nerve centerlines are applicable to vertebral landmarks in spine analysis, vascular centerlines in cardiology and neuroimaging, and airway centerlines in pulmonary assessment [36], [37]. If automatically derived labels perform comparably to hand-annotated ones—which this work tests directly through controlled comparison—the finding would have implications beyond dental CBCT.

The external validation on Philippine CBCT scans tests cross-population generalization rather than only cross-dataset robustness. Expert grading provides a clinical usability assessment that aggregate metrics alone cannot capture. The inclusion of clDice as an evaluation metric addresses a specific gap: a nerve canal prediction can post acceptable DSC while being fragmented in ways that make it unusable for implant planning or apex localization [36]. Topology must be measured directly.

Automated segmentation does not replace the clinician's ability to interpret anatomical structures—it eliminates the manual production of three-dimensional digital models required for guided surgery, orthodontic simulation, and prosthetic design, a process that currently takes 25–45 minutes per scan [5]. Reducing this to under one minute makes same-visit digital workflows economically viable in practices that cannot currently justify the time investment.

Code, pre-trained models, and the label generation tools are planned for release under permissive licensing, alongside full ablation studies and external validation results. Reproducibility is a precondition for any of this reaching routine clinical use.

1.6. Scope and Limitations (or Delimitations)

This study operates within the following boundaries:

Dataset. The primary dataset is ToothFairy3, a publicly available collection of CBCT scans with multi-class annotations covering teeth, jawbones, sinuses, and neurovascular structures [3]. To address panelist concerns regarding dataset size and to strengthen the semi-supervised learning protocol, the training pipeline supplements ToothFairy3 with unlabeled CBCT scans drawn from the STS-3D-Tooth dataset [9], which provides 148,400 unlabeled volumes alongside its 8,800 annotated scans. The

supplementation strategy is as follows: ToothFairy3's annotated scans serve as the sole source of labelled training data (with approximately 20% used as labelled and the remainder as unlabeled), while a curated subset of STS-3D-Tooth unlabeled scans is incorporated exclusively as additional unlabeled data for consistency regularization and pseudo-label generation during Mean Teacher SSL training. This approach mirrors established semi-supervised benchmarks: the STS 2023 and STS 2024 MICCAI challenges both combined small labelled sets (30–330 scans) with large unlabeled pools to train competitive models [31], [32], [33]. Importantly, the STS-3D-Tooth unlabeled scans do not require multi-class annotations and therefore introduce no additional annotation burden. The combined dataset provides both the annotation granularity required for multi-class evaluation (via ToothFairy3) and the volume of unlabeled data needed to train robust semi-supervised models. Preprocessing harmonization (resampling to common voxel spacing and intensity normalization) ensures compatibility between the two sources.

Architecture. The backbone architecture is U-Mamba2, selected because it achieved competitive results on the ToothFairy3 benchmark with clinically feasible inference times on standard GPU hardware (NVIDIA T4) [7]. The multi-task extension adds three lightweight decoder heads for primary segmentation, tooth apex landmark prediction, and nerve centerline prediction, sharing a common encoder.

Auxiliary label generation. Tooth apex landmarks are derived via three-dimensional morphological skeletonization of tooth segmentation masks. Nerve centerlines are derived via distance-transform-based medial axis thinning of canal segmentation masks. Both are computational geometry operations applied to existing annotations; they do not require additional manual labelling [36], [37].

Performance is assessed using three complementary metrics: Dice Similarity Coefficient targeting a score of 0.80 or above for volumetric overlap, 95th-percentile Hausdorff Distance targeting below 15 mm for boundary localization accuracy, and Centerline Dice targeting 0.75 or above for topological connectivity of tubular neurovascular structures. Results are disaggregated across four anatomical size tiers — Large structures covering jawbones, sinuses, and pharynx; Medium covering 32 individual teeth; Small-Pulps covering 32 pulp cavities; and Small-Tubular covering 5 neurovascular canals — to ensure that aggregate metrics do not mask failures on clinically critical small structures. Domain expert validation is conducted by two licensed dentists with CBCT experience, who independently grade a subset of test set segmentation outputs on a five-point Likert scale for clinical acceptability, with inter-rater agreement quantified via Cohen's kappa.

This study is a proof-of-concept research investigation rather than a clinical deployment system. It evaluates whether the proposed multi-task SSL framework achieves measurable performance improvements under controlled experimental conditions using publicly available benchmark data, and no web application, user interface, or clinical software is produced. Computational feasibility is demonstrated by reporting GPU requirements, model parameter count, inference latency per scan, and model file size, establishing that the approach could technically support clinical deployment on standard workstation hardware. However, translating a proof-of-concept into a deployable clinical tool would require regulatory clearance such as FDA 510(k) in the United States or equivalent pathways, prospective validation on independent clinical cohorts with pre-registered protocols, integration with picture archiving and communication systems and

dental practice management software, user acceptance testing with practicing clinicians, and ongoing monitoring for distribution shift as scanner hardware and patient populations change, all of which fall beyond the scope of an undergraduate thesis and are identified as future work.

The study carries several limitations worth acknowledging. It is limited to a single backbone architecture in U-Mamba2, and comparison with alternative architectures such as nnU-Net or Swin UNETR is not included. The automatically generated auxiliary labels depend on the quality of the primary segmentation masks and may propagate errors from imperfect ground-truth annotations, particularly for neurovascular structures where inter-annotator agreement is lowest at 72 to 78 percent Dice. The semi-supervised experimental protocol uses a fixed labelled-data ratio of 20 percent corresponding to 106 scans, and exploring multiple label fractions to characterize the full annotation-performance curve is left for future work. Cross-population generalization is not evaluated, as both datasets originate from European and Asian scanner populations, meaning performance on Filipino patient scans with potentially different dentition patterns or scanner characteristics remains unknown. Finally, the domain expert grading provides clinical usability feedback but is limited to two evaluators, which constrains the statistical power of the inter-rater agreement analysis.

Chapter 2

REVIEW OF RELATED LITERATURE & STUDIES

This chapter provides a comprehensive synthesis of the literature underpinning the present investigation into multi-task semi-supervised learning for dental CBCT segmentation. The review is organized thematically to trace the evolution of clinical needs, architectural innovations in deep learning, the emergence of multi-task learning paradigms, and label-efficient learning strategies. Each section critically examines prior research to identify patterns, contradictions, and gaps that directly motivate the research questions posed in Chapter 1. The synthesis culminates in a clear articulation of how existing knowledge converges to establish the foundation for this study's contributions.

2.1. Clinical Imperatives in AI-Driven Dental Imaging

2.1.1 Epidemiology and Clinical Burden of Dental Imaging

Cone-beam computed tomography (CBCT) has become the primary three-dimensional imaging modality for complex dental treatment planning across orthodontics, implantology, endodontics, and oral surgery [1], [2]. Each CBCT scan comprises 200 to 600 axial slices that must be systematically evaluated across three anatomical planes, and when segmentation is required for treatment planning, the manual process takes 25–45 minutes per scan for experienced practitioners [3], [5]. The capacity to interpret these scans has not scaled with imaging volume. Even among experienced oral radiologists, agreement rates for identifying anatomical structures vary considerably depending on structure size: inter-observer Dice agreement can be as high as 91–94% for large structures such as

jawbones but drops to 72–78% for small neurovascular structures like the mental foramen and accessory root canals [4].

The economic and practical implications of this interpretation bottleneck are substantial. The STS-Tooth dataset provides a concrete illustration of annotation scale: the annotation of 8,800 CBCT scans was completed by a team of 20 trained dental practitioners and verified by 6 experienced dentists, requiring approximately one year of work—and this labelled portion represented only 5.9% of the 148,400 available scans in the collection [9]. Annotating at the granularity required by challenges such as ToothFairy3, where each individual tooth, nerve canal, sinus, and bone segment must be separately delineated across over 70 classes, further increases per-scan effort [3], [18]. These annotation costs create a fundamental access problem: AI development capability concentrates in well-resourced institutions that can afford large annotation teams, while community clinics, private practices, and resource-limited settings—where AI-assisted interpretation could have the greatest clinical impact—remain excluded from these advances.

Automated deep learning-based segmentation addresses this bottleneck by reducing analysis time to under one minute per scan while maintaining agreement with expert segmentations above 90% DSC for large anatomical structures [5], [6]. Dot et al.'s DentalSegmentator, built on the nnU-Net framework and trained on over 700 CT and CBCT scans, achieved mean DSC of 92.2% on an internal test set and 94.2% on an external test set across five anatomical structure classes, demonstrating that automated tools can provide robust multiclass segmentation for routine clinical use [5]. These systems enable the production of three-dimensional digital models for guided surgery, orthodontic

simulation, and prosthetic design—workflows that would be impractical at scale if every model required 25–45 minutes of manual delineation.

2.1.2 Anatomical Complexity of Maxillofacial Structures

The maxillofacial region presents distinctive challenges for automated image analysis due to the number of structures, their size variability, and inter-patient anatomical differences [1]. Adult dentition alone comprises up to 32 permanent teeth divided into four functional categories—incisors, canines, premolars, and molars—each with distinct crown geometry, root count, and internal pulp architecture. Clinical utility demands not merely detecting that a tooth is present, but identifying which specific tooth it is and localizing its internal landmarks: the anatomical apex (root tip) must be located within one to two millimetres for endodontic procedures to succeed safely [16].

Patient-specific variation compounds this challenge across three dimensions documented by Rizwan et al. [16] in a prospective study of 320 paediatric CBCT scans. Developmental variability means that children in mixed dentition may present up to 52 dental elements simultaneously, with root maturation ranging from absent to fully formed. Pathological change further disrupts expected morphology: caries, periapical infections, and developmental anomalies such as supernumerary or ectopic teeth alter the structures that segmentation models expect to find. Prior dental treatment introduces additional complications: metallic restorations produce beam-hardening artefacts—bright streaks and dark shadows that radiate outward from the restoration—obscuring adjacent anatomy and rendering boundaries between structures indistinct even for experienced clinicians [16], [21].

Neurovascular anatomy presents the most clinically critical segmentation targets. The inferior alveolar canal, which houses the nerve supplying sensation to the lower teeth and lip, follows a path through the mandible with substantial variability: canal trajectory can deviate more than two millimetres from its expected position, may bifurcate into multiple channels, and often exhibits poorly defined borders in regions of low bone density [17]. Kanneppady et al. [17] documented through retrospective analysis of 120 implant surgery cases that inadequate identification of this canal contributed to intraoperative nerve injury, with some cases resulting in permanent sensory deficits—outcomes carrying both patient safety and medico-legal consequences. The mental foramen and incisive canal occupy less than 0.1% of total scan volume while exhibiting positional variation exceeding three millimetres across patients [17]. Their low radiographic contrast relative to surrounding trabecular bone makes reliable delineation difficult even under expert review.

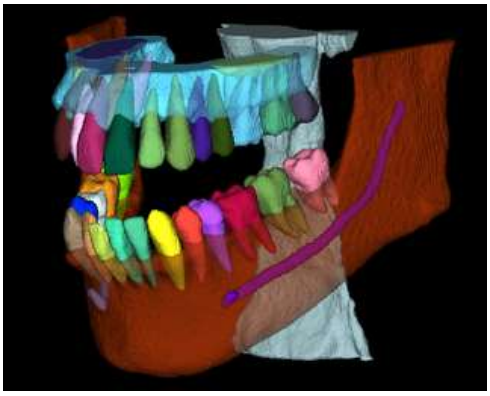

Figure 1. Anatomical Complexity of the Maxillofacial Region. Reprinted from [40].

From a computational perspective, these characteristics translate into specific requirements. Models must handle over 40 anatomical classes with extreme volume imbalance: jawbones may occupy 30–40% of scan volume while critical neurovascular

structures together constitute less than 0.1% [3], [7]. This imbalance causes naively trained models to systematically under-represent clinically important rare structures. The model must also reason spatially across the full extent of the jaw—connecting structures that are anatomically related but physically separated by 20–30 mm—while simultaneously resolving sub-millimetre boundaries between adjacent teeth [17]. These requirements directly inform this study's choice of architecture (U-Mamba2 for efficient long-range modelling), loss function design (weighted losses for class imbalance), and auxiliary task selection (apex landmarks, nerve centerlines).

2.1.3 The Diagnostic Superiority of CBCT Over Two-Dimensional Radiography

The transition from two-dimensional intraoral and panoramic radiography to three-dimensional CBCT represents one of the most significant technological advances in dental diagnostics [1], [2]. Rizwan et al. [16] conducted a controlled comparison involving 320 paediatric patients, reporting that CBCT detected 88.2% of periapical lesions compared to 64.7% for intraoral radiographs ($p < 0.001$). The diagnostic advantage was most pronounced for posterior teeth, where anatomical superimposition in two-dimensional projections had previously obscured pathology. Kanneppady et al. [17] documented through retrospective analysis that CBCT identified 34% more periapical lesions than periapical films in 120 adult cases, attributing the improvement to three CBCT advantages: elimination of anatomical superimposition, detection of early-stage bone resorption below one millimetre in diameter, and visualization of buccal-lingual cortical plate perforation undetectable in two-dimensional projections.

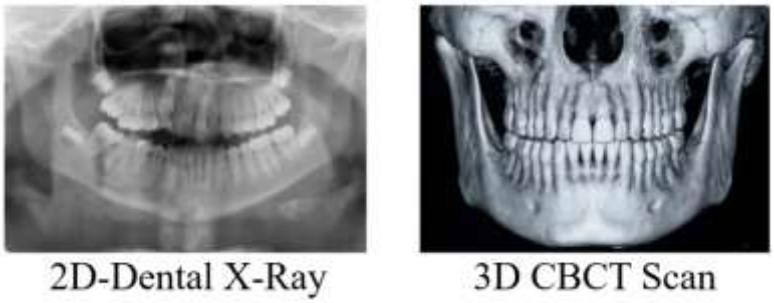

Figure 2. 2D Radiography vs CBCT Scan Diagnostic Comparison. Reprinted from [41]

CBCT employs a cone-shaped X-ray beam and flat-panel detector that acquires volumetric data in a single rotation, producing isotropic voxels of 0.08–0.4 mm—higher spatial resolution than conventional multi-slice CT (0.5–1.0 mm) while delivering lower radiation doses [2]. This resolution enables visualization of structures below one millimetre in diameter, including accessory root canals, early carious lesions, and fine trabecular bone architecture [1], [2]. Almufarrij and Alobaid [2] documented clinical applications spanning multiple specialties: in implant dentistry, CBCT enables three-dimensional assessment of bone volume and proximity to critical structures; in orthodontics, it facilitates evaluation of impacted teeth and root resorption monitoring; in endodontics, it identifies complex root canal anatomy undetectable on periapical radiographs.

CBCT's diagnostic capability introduces interpretation demands that create the clinical imperative for automated segmentation. Each scan's hundreds of axial slices require systematic review, three-dimensional reconstruction of spatial relationships, and integration of findings across multiple anatomical planes. This cognitive load, combined with the need to identify subtle pathology occupying less than 0.1% of voxel volume,

creates risk of oversight—particularly for neurovascular structures and early periapical lesions that lack conspicuous visual features [16], [17]. Automated segmentation tools that can rapidly and consistently delineate all relevant anatomical structures serve as a practical solution to these interpretation demands.

2.2. Deep Learning Architectures for Medical Image Segmentation

The evolution of deep learning architectures for volumetric medical image segmentation reflects an ongoing tension between three competing objectives: capturing long-range spatial dependencies essential for anatomical reasoning, maintaining computational efficiency suitable for clinical deployment, and learning effectively from limited annotated data. This section traces the architectural progression from convolutional neural networks through transformer-based approaches to state space models, establishing U-Mamba2 as the architectural foundation for this study's multi-task semi-supervised framework.

2.2.1 Convolutional Neural Networks: The Established Foundation

Ronneberger et al.'s U-Net architecture established the encoder-decoder framework with skip connections as the standard design for biomedical image segmentation [22]. The architecture's symmetric U-shaped design comprises a contracting path that captures contextual information at progressively lower spatial resolutions, an expanding path that

recovers precise localization through upsampling, and skip connections that preserve spatial details lost during downsampling.

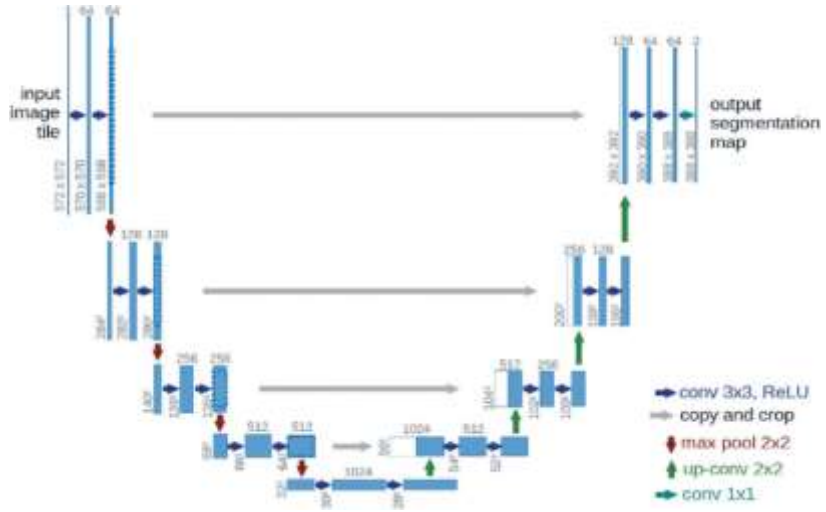

Figure 3. U-Net Encoder-decoder Architecture with Skip Connections. Retreived from [43]

Three-dimensional extensions enabled direct volumetric processing of CBCT data, where inter-slice context informs anatomical delineation and three-dimensional relationships must be preserved. However, memory requirements scale cubically with spatial dimensions, and the local receptive field of convolutional operations constrains modelling of long-range spatial dependencies.

Isensee et al.'s nnU-Net framework [22] demonstrated that systematic, self-configuring hyperparameter optimization—automatically determining patch sizes, batch sizes, network topology, and augmentation strategies based on dataset characteristics—could achieve competitive performance across diverse tasks without manual architectural engineering. The subsequent nnU-Net v2 [23] reinforced this finding, showing that well-

configured CNNs remain competitive with newer architectures when rigorously trained. On the ToothFairy challenge series, nnU-Net has consistently served as the primary CNN baseline against which newer architectures are benchmarked [3], [7].

Chen et al.'s [13] multi-task fully convolutional network demonstrated that CNN architectures could support multi-task learning for dental CBCT by simultaneously predicting tooth region probabilities and surface probabilities. Their architecture achieved a 3.8% DSC improvement over single-task baselines through the auxiliary surface prediction task, which enforced explicit boundary awareness in the shared encoder features [13]. This result established that carefully chosen auxiliary tasks can enhance primary segmentation objectives through shared feature learning—a principle central to the present study. However, their method required separate manual annotation for surface labels, increasing per-scan annotation time by 33–50%.

Despite these advances, convolutional architectures face an inherent limitation: the effective receptive field grows slowly with network depth and remains substantially smaller than theoretical limits [28]. For dental anatomy where spatial relationships between distant structures (a tooth apex and the inferior alveolar canal 20–30 mm below it) inform segmentation decisions, this local bias becomes a constraint that motivated exploration of architectures capable of explicit global modelling.

2.2.2 Transformers: Global Attention at Computational Cost

Vision Transformers addressed the receptive field limitation of CNNs by treating images as sequences of patches and computing pairwise interactions between all patch embeddings through self-attention mechanisms. This enables explicit global dependency

modelling from the first layer, allowing arbitrary spatial relationships to be captured without relying on hierarchical stacking of local operations.

Hatamizadeh et al.'s UNETR [24] pioneered transformer application to three-dimensional medical segmentation by employing a Vision Transformer encoder that divides the input volume into non-overlapping three-dimensional patches, processes them as a flattened sequence through self-attention layers, and then feeds the encoded features into CNN-based decoder blocks with skip connections. However, the quadratic computational complexity of self-attention poses challenges for high-resolution CBCT: for a typical 512×512×148 scan, the number of pairwise attention computations becomes prohibitive for clinical deployment on standard workstation hardware.

Hierarchical transformer architectures emerged to address this scalability issue. Hatamizadeh et al.'s Swin UNETR [25] adapted the shifted-window self-attention mechanism for medical imaging, computing attention within non-overlapping local windows and shifting window positions between layers to introduce cross-window connections. This approach reduces computational cost to linear with respect to input size while maintaining the capacity for global modelling through hierarchical propagation. On dental CBCT data, transformer-based approaches demonstrated strong performance on large structures but reduced accuracy for small neurovascular anatomies [7]—suggesting

that windowed attention may sacrifice the fine-grained global context necessary for resolving structures that occupy less than 0.1% of scan volume.

Table 1. Architecture Comparison on Dental CBCT Segmentation

Architecture	Core Module	ToothFairy3 Dice	Complexity
nnU-Net [7]	CNN	Competitive baseline	O(N)
UNETR [26]	ViT	Lower on CBCT	$O(N^2)$
Swin UNETR [27]	Swin-T	Strong on large	O(N)
U-Mamba2 [16]	Mamba2	Top-3 TF3 (0.792 test Dice)	O(L)

2.2.3 State Space Models: Efficient Long-Range Modeling

Gu and Dao's Mamba architecture [26] reconciled global modelling capability with computational efficiency by introducing input-dependent selectivity to structured state space models. Building on the S4 framework, Mamba enables the model to dynamically determine which information to propagate through the state space based on input content, capturing long-range dependencies at linear computational cost with respect to sequence length. This provides a middle ground between CNNs' fixed local receptive fields and transformers' complete but computationally expensive global attention.

Dao and Gu's Mamba2 [27] further improved upon this through the structured state space duality (SSD) framework, which enforces stronger constraints on the hidden space structure and reformulates the selective scan operation to enable tensor and sequence parallelism. These modifications provide substantial speedup without compromising modelling capacity—an advantage relevant to clinical deployment where inference time

directly affects workflow integration. The broader impact of Mamba architectures in medical imaging has been documented in several survey papers: Bansal et al. [38] and Heidari et al. [39] provide comprehensive reviews of Mamba applications across classification, segmentation, and reconstruction tasks in medical imaging.

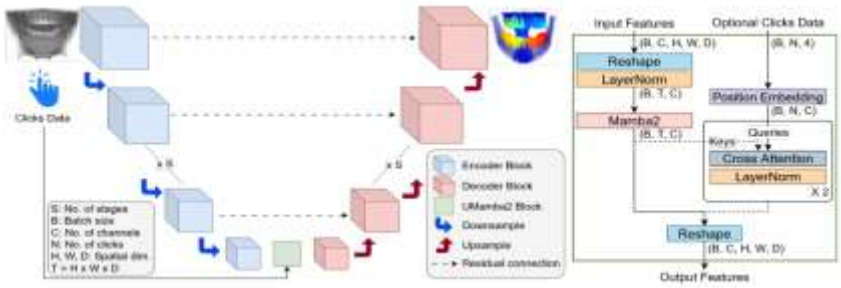

Figure 4. U-Mamba2 model Architecture. Retrieved from [16]

Tan et al.'s U-Mamba2 [7], developed for the ToothFairy3 dental CBCT segmentation challenge, integrates Mamba2 blocks into the bottleneck stage of a seven-stage U-Net encoder-decoder framework, with CNN-based residual blocks handling encoding and decoding at higher spatial resolutions. This hybrid design leverages CNNs' efficiency for local feature extraction while employing Mamba2's selective state space mechanism for long-range dependency modelling at the bottleneck where reduced spatial resolution makes sequence length manageable. On the ToothFairy3 hidden test set, U-Mamba2 achieved a mean Dice of 0.792 and HD95 of 93.19, securing a top-three position in Task 1 of the challenge [7].

Several domain-specific design choices distinguish U-Mamba2 from generic segmentation architectures. Label smoothing for anatomically related structures (left-right counterparts, neighbouring teeth within the same arch) guides the model's recognition of semantic similarities. Weighted loss functions applying elevated weights to structures

occupying less than 0.1% of voxels (incisive nerves, mental foramina, lingual foramen) compensate for extreme class imbalance. Bilateral symmetry handling through label-swapping during mirroring augmentation—exchanging labels for left-right anatomical pairs when horizontally flipping scans—exploits maxillofacial anatomy's inherent symmetry to expand training data diversity without creating anatomically impossible configurations [7]. These domain-aware choices are directly relevant to the present study, which extends U-Mamba2 with multi-task decoder heads while preserving these design principles.

2.3. Multi-Task Learning in Medical Image Analysis

Multi-task learning (MTL) trains models to predict related outputs jointly within unified architectures. By forcing a shared encoder to satisfy multiple objectives simultaneously, MTL encourages learning richer and more generalizable feature representations than single-task training. This section reviews MTL's theoretical basis, its empirical benefits in dental segmentation, the problem of negative transfer, the critical limitation of annotation multiplication, and the methodological foundation for automatic auxiliary label generation.

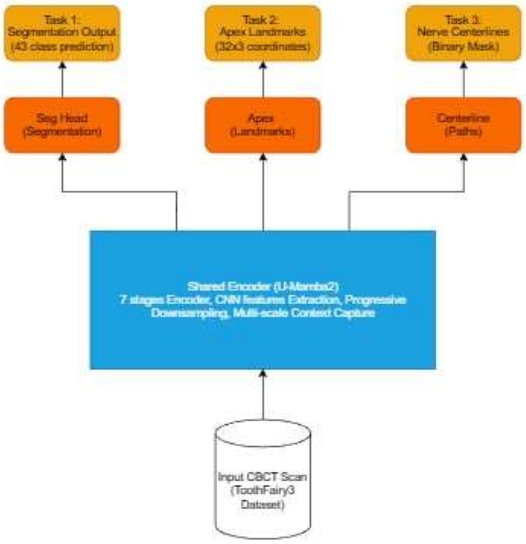

Figure 5. Multi-Task Framework

2.3.1. Evidence for Multi-Task Benefits in Dental Segmentation

Vandenhende et al.'s [12] comprehensive survey of MTL for dense prediction tasks established that joint training acts as an implicit regularizer: by forcing the shared encoder to learn features useful across multiple objectives, MTL reduces overfitting and improves generalization relative to single-task learning, particularly when training data is limited. Crawshaw [34] extended this analysis, documenting that MTL's regularization effect is strongest when tasks share underlying structure and when labelled data is scarce—precisely the conditions that characterize dental CBCT segmentation. Li et al. [35] provided theoretical grounding by proving that multi-task training in overparameterized neural networks induces implicit regularization that narrows the hypothesis space, reducing the effective model complexity and thereby lowering the risk of overfitting on small datasets.

Empirical evidence in dental CBCT segmentation confirms these theoretical benefits. Chen et al. [13] demonstrated that jointly predicting tooth region probabilities and surface probabilities improved DSC by 3.8% over single-task baselines. The authors attributed this gain to the surface prediction task's role in enforcing explicit boundary awareness: gradients from the surface probability loss encouraged the shared encoder to extract features sensitive to tooth-background transitions, improving primary segmentation boundary precision. Wang et al. [14] achieved 4.2% DSC gains through hierarchical multi-task learning that coupled semantic segmentation with instance segmentation and root canal identification, finding that the disproportionately large improvement for the most difficult task (canal segmentation) suggested auxiliary tasks are particularly beneficial when primary task difficulty is high. Duan et al. [15] demonstrated 5.3% improvement by pairing canal segmentation with boundary refinement, confirming that even simple auxiliary tasks focusing on different anatomical aspects (volume versus boundary) provide measurable benefits.

2.3.2. Task Selection and the Risk of Negative Transfer

Not all task combinations produce benefits. Negative transfer occurs when tasks require contradictory feature representations or when one task dominates gradient contributions, suppressing learning for others [12], [34]. Vandenhende et al. [12] documented that task complementarity—the degree to which tasks capture different but related aspects of the data—is the primary determinant of MTL success. When auxiliary tasks encode information that is both different from and complementary to the primary task (for example, volumetric segmentation paired with boundary detection, or coarse

classification paired with fine-grained localization), the shared encoder is pushed to learn richer features than either task alone would demand.

For the present study, auxiliary task selection is guided by clinical relevance and anatomical complementarity. Tooth apex landmark prediction captures geometric point-localization information that complements volumetric segmentation by enforcing awareness of root tip positions—critical for endodontic planning. Nerve centerline prediction captures tubular topology that complements volumetric canal segmentation by enforcing awareness of structural connectivity—critical for implant planning where a fragmented canal prediction is clinically useless even if it achieves acceptable DSC. Both auxiliary tasks encode information that is structurally distinct from (but anatomically related to) the primary segmentation masks, minimizing the risk of negative transfer while maximizing the complementary learning signal.

2.3.3. The Critical Limitation: Annotation Multiplication

Despite MTL's demonstrated efficacy, a fundamental limitation constrains its applicability in annotation-scarce settings: existing approaches require separate manual annotation for each task, multiplying rather than reducing annotation burden. Chen et al.'s [13] dual-task network required radiologists to label both tooth volumes and tooth surfaces, increasing annotation time by 33–50% per scan. Wang et al.'s [14] three-task framework necessitated hierarchical annotation of semantic regions, instance boundaries, and canal paths, effectively tripling the annotation effort. Duan et al.'s [15] boundary refinement approach similarly required dedicated boundary annotations alongside volumetric canal labels.

This annotation multiplication negates MTL's efficiency premise in the very scenarios where semi-supervised learning is most needed. If acquiring labelled scans for single-task training is already a bottleneck, requiring multiple sets of task-specific annotations per scan becomes infeasible even for well-resourced institutions. More critically, this limitation concentrates MTL benefits in settings with abundant annotation resources—the opposite of where improved data efficiency is most needed. None of the reviewed dental segmentation studies [13], [14], [15] addressed this limitation through automatic auxiliary label generation from existing primary task labels—the gap that this research directly targets.

2.3.4 Automatic Auxiliary Label Generation: Methodological Foundation

Morphological skeletonization and tubular structure thinning are well-established techniques in medical image analysis, with applications spanning vessel centerline extraction, airway tree analysis, and spine segmentation [36], [37]. Their specific application to dental CBCT multi-task learning within semi-supervised training frameworks, however, remains unexplored.

Lee et al.'s [37] foundational work on three-dimensional medial surface and axis thinning algorithms established the mathematical framework for skeletonization: iterative boundary erosion that preserves topology while reducing objects to one-voxel-wide centerlines, with endpoints corresponding to terminal branches. For tooth segmentation masks, applying skeletonization yields medial axes with endpoints at crown tips and root apices—the latter representing clinically critical landmarks. The approach is computationally efficient (typically seconds per structure on modern hardware) and robust to boundary noise (skeleton topology remains stable under small boundary perturbations),

properties essential for automatic label generation where slight variations in segmentation mask boundaries should not produce drastically different landmark predictions.

Nerve centerline extraction follows analogous principles using distance transform and morphological thinning. The Euclidean distance transform assigns each foreground voxel its distance to the nearest background voxel; local maxima of this distance field correspond to centerline positions; and subsequent thinning reduces the result to one-voxel width. Shit et al.'s clDice work [36] demonstrated that centerline-based representations are both computationally tractable and clinically meaningful for tubular structures—properties that transfer directly to dental nerve canal analysis. The present study hypothesizes that if auxiliary labels generated by these morphological operations can recover even part of the performance gains associated with hand-annotated MTL tasks, multi-task learning becomes practically accessible without annotation cost increase.

2.4. Semi-Supervised Learning for Medical Image Segmentation

Semi-supervised learning (SSL) enables models to learn from both small sets of labelled examples and large pools of unlabeled data, addressing the annotation bottleneck that limits supervised learning in medical imaging. This section synthesizes SSL's theoretical foundations, reviews the Mean Teacher framework, examines recent advances in dental CBCT semi-supervised segmentation, and discusses U-Mamba2-SSL's achievements and remaining limitations.

2.4.1. Theoretical Foundations and the Consistency Assumption

SSL's theoretical foundation rests on two core assumptions. The smoothness assumption posits that nearby points in feature space should receive similar predictions: if two unlabeled samples have similar representations, they likely belong to the same

semantic category. The cluster assumption posits that data naturally clusters by semantic category with low-density regions between classes, so decision boundaries should traverse low-density regions rather than cutting through data clusters [10], [11]. In dental CBCT segmentation, these assumptions align with anatomical consistency: similar-appearing tissues across different patients typically correspond to the same anatomical structures—enamel exhibits characteristic high intensity, mandibular cortical bone shows consistent texture patterns—making SSL particularly effective for tasks with strong anatomical priors.

Yang et al.'s [10] comprehensive survey of deep semi-supervised learning identified consistency regularization as the dominant paradigm in medical image analysis, operationalizing the smoothness assumption by enforcing prediction invariance under input perturbations. Jiao et al.'s [11] survey focused specifically on SSL for medical image segmentation, documenting that consistency-based methods outperform purely pseudo-label-based approaches when labelled data is extremely scarce (below 10% of the dataset), while hybrid approaches combining consistency regularization with pseudo-labelling achieve the best performance across moderate label ratios (10–30%).

2.4.2. Mean Teacher Framework

The Mean Teacher framework [28] maintains two models: a student model updated via gradient descent and a teacher model updated as an exponential moving average (EMA) of student weights. The teacher provides pseudo-targets for the student's consistency loss on unlabeled data: the student is trained to produce predictions on strongly augmented unlabeled samples that match the teacher's predictions on weakly augmented versions of the same samples. The EMA update mechanism produces temporally smoothed predictions that are more stable than the student's direct outputs, providing higher-quality supervision

than direct self-ensembling. This framework has become the de facto backbone for semi-supervised medical image segmentation [10], [11].

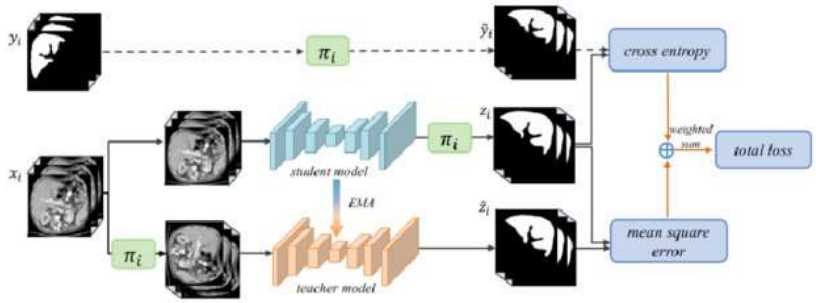

Figure 6. Mean Teacher Pipeline for Medical Image Analysis

Recent improvements extend the Mean Teacher paradigm in several directions relevant to this study. Li et al. [29] introduced self-loop uncertainty estimation for pseudo-label quality assessment, enabling the model to weight reliable pseudo-labels more heavily during training. Lyu et al. [30] proposed region-aware instructive learning specifically for semi-supervised dental segmentation, incorporating spatial priors about tooth arrangement to improve pseudo-label accuracy in crowded dental arches. Cross-pseudo-supervision approaches [28] enforce consistency between two independently trained networks, reducing the risk of confirmation bias where a single model's errors propagate uncorrected through its own pseudo-labels.

2.4.3. Semi-Supervised Dental CBCT Segmentation: Benchmarks and State of the Art

The MICCAI STS challenge series has served as the primary benchmark for semi-supervised dental segmentation. The STS 2023 challenge [31] provided over 580 CBCT volumes with partial annotations, demonstrating that combining approximately 30 labelled scans with several hundred unlabeled ones could yield clinically acceptable tooth segmentation. A retrospective analysis of challenge methods [32] found that entropy-guided Mean Teacher models and multi-stage self-training strategies produced the most consistent improvements across different label ratios. The STS 2024 challenge [33] extended evaluation to instance-level segmentation across both paediatric and adult dentitions, providing a dataset of 2,380 OPG images and 330 CBCT scans. The winning semi-supervised method improved Instance Dice by 61 percentage points over a fully supervised nnU-Net baseline trained only on labelled data, confirming SSL's substantial benefit for complex segmentation tasks where labelled data is scarce [33].

Tan et al.'s U-Mamba2-SSL [8] represents the current state of the art for semi-supervised dental segmentation. Working within the STSR 2025 challenge for tooth and pulp segmentation, U-Mamba2-SSL achieved 91.7% DSC using only 8% labelled data through a three-stage strategy: (1) self-supervised pre-training via a disruptive autoencoder that learns robust feature representations from unlabeled data, (2) consistency regularization using the Mean Teacher framework with strong and weak augmentations, and (3) pseudo-labelling where confident teacher predictions on unlabeled data are incorporated as training labels for the student. This result secured first place in the

challenge and demonstrated that properly configured SSL can match or exceed fully supervised baselines trained on substantially more labelled data [8].

However, U-Mamba2-SSL [8] optimized only the primary segmentation task without incorporating auxiliary anatomical priors through multi-task learning. Whether U-Mamba2's architectural properties, which benefit fully supervised and single-task SSL scenarios, translate to multi-task SSL settings involving automatically generated auxiliary labels remains empirically unvalidated. Multi-task SSL introduces unique optimization challenges: consistency regularization must hold across multiple task outputs simultaneously, and auxiliary task pseudo-labels may exhibit different noise characteristics than primary segmentation pseudo-labels, potentially destabilizing training if not properly balanced. The present study directly investigates this gap.

2.5. Evaluation Metrics for Dental CBCT Segmentation

Standard segmentation evaluation relies on the Dice Similarity Coefficient (DSC), which measures volumetric overlap between predicted and ground-truth masks, and the 95th-percentile Hausdorff Distance (HD95), which measures the worst-case boundary localization error after excluding 5% of the most extreme outliers. Both metrics are well-established and widely used across medical image segmentation benchmarks [3], [7], [22]. However, for thin tubular structures such as the inferior alveolar canal, incisive nerve, and mental foramen canal, these metrics have important limitations.

DSC measures volumetric overlap but does not capture whether a tubular structure remains topologically connected. A predicted nerve canal split into two disconnected segments can still achieve a reasonable DSC if the total volume of the predicted voxels approximates the ground truth, yet such a fragmented prediction would be clinically useless

for implant planning—a dentist cannot plan around a nerve path that appears to terminate mid-mandible. Similarly, HD95 measures boundary distance but does not penalize gaps in connectivity: a prediction with a two-millimetre gap in the middle of a canal and perfect boundary alignment elsewhere could achieve acceptable HD95 while being unsafe for surgical guidance.

The centerline Dice (clDice) metric, introduced by Shit et al. [36], addresses this limitation by evaluating overlap between the morphological skeletons (centerlines) of the predicted and ground-truth structures. clDice first extracts one-voxel-wide centerlines from both the prediction and the ground truth through morphological skeletonization, then computes the fraction of each skeleton that lies within the other's volumetric mask. The harmonic mean of these two fractions (topology precision and topology sensitivity) yields clDice. Critically, Shit et al. proved that clDice = 1.0 guarantees topology preservation up to homotopy equivalence for binary segmentation in both two and three dimensions [36]. clDice is now standard in vascular segmentation and airway analysis but has not been applied to dental CBCT evaluation—an omission this study addresses by reporting clDice for all tubular neurovascular structures alongside DSC and HD95.

The present study therefore adopts a three-metric evaluation framework: DSC for overall volumetric agreement, HD95 for boundary localization accuracy, and clDice for topological connectivity of tubular structures. Results are disaggregated across three anatomical size categories—large (jawbones, sinuses), medium (individual teeth), and small (neurovascular canals, foramina)—to ensure that aggregate metrics do not mask failures on clinically critical small structures.

2.6. Synthesis and Research Gaps

The reviewed literature establishes that dental CBCT segmentation faces converging challenges. CBCT's diagnostic superiority over two-dimensional radiography [16], [17] positions it as the primary advanced imaging modality, yet interpretation and segmentation demands consume 25–45 minutes per scan for experienced practitioners [3], [5]. Annotation costs—illustrated by the year-long effort required to label 5.9% of the STS-Tooth collection [9]—create economic barriers that concentrate AI development in well-resourced institutions. Anatomical complexity demands precise localization of over 40 structures spanning a volume range from 30–40% (jawbones) to less than 0.1% (neurovascular canals) of the scan, with substantial inter-patient variability [3], [17].

Architecturally, the progression from CNNs [22], [23] through transformers [24], [25] to state space models [26], [27] has delivered increasing long-range dependency modelling effectiveness with decreasing computational cost. U-Mamba2 secured a top-three position in the ToothFairy3 challenge with clinically feasible inference times [7], establishing technical feasibility. However, U-Mamba2's effectiveness in semi-supervised scenarios with substantially reduced label ratios (20%) and multi-task extensions remains untested.

Multi-task learning consistently improves segmentation—by 3.8% [13], 4.2% [14], and 5.3% [15] DSC in dental applications—when auxiliary tasks capture complementary anatomical information, but existing approaches require separate manual annotation for each task, multiplying annotation burden. Semi-supervised learning achieves dramatic gains under annotation scarcity (the STS 2024 winning method improved Instance Dice by 61 percentage points over a supervised baseline [33]; U-Mamba2-SSL reached 91.7% DSC

with 8% labels [8]), yet SSL and MTL for dental CBCT have developed in parallel rather than in combination.

Three critical gaps motivate this research:

Gap 1: No integration of SSL and MTL for dental CBCT. SSL studies [8], [31], [32], [33] focus on single-task segmentation without anatomical priors. MTL studies [13], [14], [15] operate in fully supervised settings without leveraging unlabeled data. The interactions between these two paradigms—including challenges of multi-output consistency regularization and auxiliary pseudo-label noise—remain unexplored.

Gap 2: No automatic auxiliary label generation for dental MTL. All existing dental MTL approaches require separate manual annotation for each auxiliary task, negating efficiency in annotation-scarce scenarios. Whether morphological operations (skeletonization for apex landmarks [37], distance-transform thinning for nerve centerlines [36]) can generate auxiliary labels that recover part of hand-annotated MTL's performance gains has not been tested.

Gap 3: Inadequate evaluation of tubular structure connectivity. Standard metrics (DSC, HD95) do not capture whether thin neurovascular predictions are topologically intact. clDice [36] is routine in vascular segmentation but has not been applied to dental CBCT, leaving a measurement gap that may allow clinically unsafe fragmented predictions to appear acceptable by standard metrics.

This synthesis reveals that the constituent elements for annotation-efficient dental CBCT segmentation exist in isolation: effective SSL methods (Mean Teacher [28]), beneficial MTL paradigms [13], [14], [15], efficient architectures (U-Mamba2 [7]), topology-preserving metrics (clDice [36]), and automatic label generation techniques [36],

[37]. Their integration into a unified framework—combining U-Mamba2's efficiency, automatically generated auxiliary labels, and semi-supervised protocols trained on ToothFairy3 supplemented with STS-3D-Tooth unlabeled data [9]—while systematically evaluating performance with topology-aware metrics and external validation, represents the novel contribution this research pursues.

Chapter 3

METHODOLOGY

Commented [JM1]: lock in

This chapter describes the research design, dataset strategy, preprocessing pipeline, automatic auxiliary label generation procedures, model architecture, semi-supervised training protocol, evaluation metrics, domain expert validation, deployment considerations, timeline, and ethical considerations. The methodology is designed to be completed within a 12-week timeframe using publicly available datasets and cloud-based computing resources.

3.1 Conceptual Framework

The conceptual framework integrates three methodological pillars: (1) U-Mamba2 as the backbone architecture for efficient multi-anatomy CBCT segmentation, (2) Mean Teacher semi-supervised learning to leverage unlabeled data alongside a small labelled set, and (3) multi-task learning with automatically generated auxiliary labels to enrich feature representations at zero incremental annotation cost. The framework's central hypothesis is that automatically derived tooth apex landmarks and nerve centerlines, when used as auxiliary training objectives within a semi-supervised protocol, improve segmentation performance beyond a single-task SSL baseline—particularly for small, clinically critical neurovascular structures.

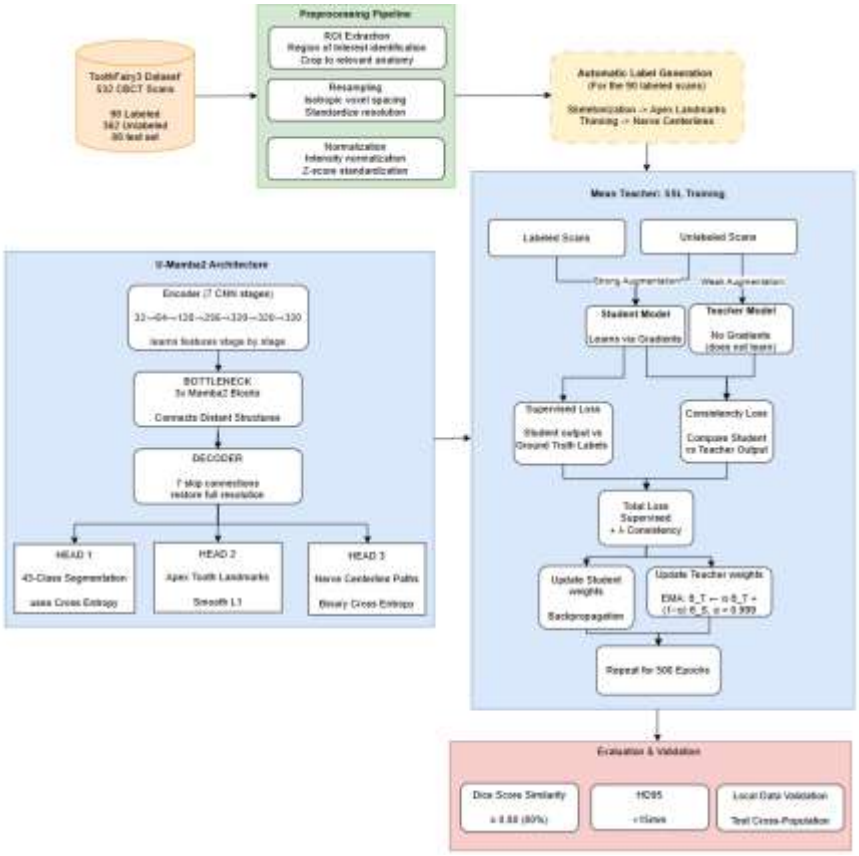

Figure 7. Conceptual Framework Diagram

3.2. Research Design

This study employs a controlled comparative experimental design to test whether automatically generated auxiliary task labels improve primary segmentation performance in a semi-supervised learning setting. The experimental logic is straightforward: if all training conditions are held constant and only the presence of auxiliary tasks varies between configurations, any observed performance difference can be attributed directly to

the auxiliary task supervision. Two configurations are therefore defined and trained under identical conditions. Configuration 1, the single-task baseline, trains U-Mamba2 with a single decoder head predicting only the primary 77-class anatomical segmentation. Configuration 2, the multi-task full model trains U-Mamba2 with three decoder heads predicting the primary segmentation simultaneously with tooth apex landmark coordinates and nerve centerline masks. Both configurations employ Mean Teacher semi-supervised learning [15] with 20% labeled data, identical hyperparameters, and a 500-epoch training duration of approximately six days per configuration on an NVIDIA A100 GPU, ensuring that any observed performance difference between the two is attributable solely to the presence of automatic auxiliary task supervision rather than any difference in architecture, data, or training procedure.

Table 2. Experimental Configuration Summary

Aspect	Config A (Baseline)	Config B (Proposed)
Architecture	U-Mamba2 (77-class)	U-Mamba2 (77-class)
SSL Framework	Mean Teacher	Mean Teacher
Segmentation Classes	77 (ToothFairy3)	77 (ToothFairy3)
Task Heads	1 (Segmentation only)	3 (Seg + Apex + Centerline)
Auxiliary Labels	None	Auto-generated via morphological operations
Labelled Data	106 scans (20% of TF3)	106 scans (20% of TF3)

3.3. Dataset and Data Preparation

3.3.1 ToothFairy3 Dataset

The primary dataset is ToothFairy3 [43], the most comprehensive publicly available benchmark for multi-structure dental CBCT segmentation, released as part of the ODIN2025 challenge cluster at MICCAI 2025 [3]. Its 532 CBCT volumes are organized into three subsets acquired across two scanner types. Set A comprises 417 volumes from the first scanner, sharing the same field of view as the original ToothFairy and ToothFairy2 datasets (prefixed "P"); Set B contributes 63 volumes from the same scanner but with a broader field of view that captures complete upper teeth segmentations (prefixed "F"); and Set C provides 52 volumes from a different scanner introduced in ToothFairy3, adding cross-scanner diversity to the collection (prefixed "S").

All 532 volumes carry dense voxel-wise annotations across 77 anatomical classes following an extended FDI World Dental Federation notation [43]. These span two jawbone classes (mandible and maxilla), two inferior alveolar canals, two maxillary sinuses, one pharynx, three dental restoration classes (bridge, crown, implant), 32 individual tooth classes (FDI 11–48), two mandibular incisive canals, one lingual canal, and 32 tooth pulp cavity classes — one per tooth. Data are stored in NIfTI format (.nii.gz) following the nnU-Net dataset structure.

ToothFairy3 is selected because it is the only publicly available dataset providing annotations at this level of granularity. The expansion from 42 classes in ToothFairy2 to 77 — adding pulp cavities, incisive canals, and the lingual canal — makes the segmentation task substantially more demanding and directly relevant to endodontic and surgical planning. No other public dataset offers comparable annotation depth.

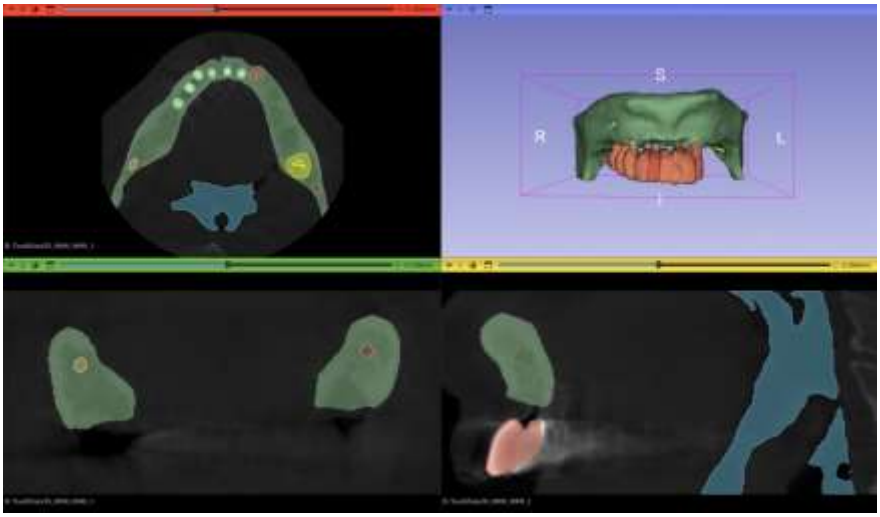

Figure 8. CBCT Scan in different views

3.3.2 STS-3D-Tooth Dataset

To strengthen the semi-supervised learning protocol, the training pipeline supplements ToothFairy3 with 600 unlabeled CBCT scans drawn from the STS-3D-Tooth dataset [44]. STS-3D-Tooth contains 148,400 CBCT scans, of which 8,800 carry tooth-level annotations and the remainder are unlabeled. The annotation work was completed by 20 trained dental practitioners and verified by 6 experienced dentists over approximately one year [44], [9].

The use of STS-3D-Tooth as a supplementary unlabeled source follows directly from Tan et al.'s U-Mamba2 [7], which used 371 unlabeled STS-3D-Tooth scans for self-supervised pre-training alongside the 532 ToothFairy3 scans. The present study extends this to 600 unlabeled scans because Mean Teacher consistency regularization benefits from greater data diversity than one-time pre-training: consistency loss is computed on every unlabeled sample in every training epoch, making a larger unlabeled pool directly

beneficial. The 600-scan subset is curated from the full STS-3D-Tooth collection through quality filtering for: (a) scan completeness (full mandible visible), (b) absence of severe motion artifacts, and (c) adequate field-of-view coverage. Only the unlabeled portion is used; no STS-3D-Tooth annotations are incorporated, as STS-3D-Tooth's tooth-only labels do not match ToothFairy3's 77-class annotation schema.

The combined dataset of 1,132 scans (532 ToothFairy3 + 600 STS-3D-Tooth) exceeds the dataset sizes used by top-performing methods in the MICCAI STS 2023 challenge (~580 CBCT volumes [31], [32]) and the STS 2024 challenge (~330 CBCT volumes [33]), and is directly comparable to the 903 total scans (532 + 371) used by Tan et al. [7] in U-Mamba2 development. The supplementation introduces no additional annotation cost.

3.3.3 Dataset Splitting

The 532 ToothFairy3 annotated scans are divided into non-overlapping subsets. The 600 STS-3D-Tooth scans are used exclusively as additional unlabeled training data.

Table 3. Dataset Split Summary

Subset	Source	Scans	Labels	Role in Training
Labelled Train	TF3	106	77-class GT	Supervised loss (ground truth + auto-generated aux labels)
Unlabeled Train	TF3	326	Withheld	Mean Teacher consistency regularization
Unlabeled Train	STS-3D	600	None	Additional SSL unlabeled pool

Validation	TF3	50	77-class GT	Hyperparameter tuning, early stopping
Test	TF3	50	77-class GT	Final metric reporting (used once)
TOTAL TRAINING	Both	1,032		106 labelled + 926 unlabeled

3.3.4 ToothFairy3 Anatomical Region Classification and Clinical Significance

The 77 annotated classes in ToothFairy3 [43] span six functional categories of maxillofacial anatomy, each with distinct clinical significance, volumetric proportion, and segmentation difficulty. Understanding these categories is essential for interpreting disaggregated evaluation results and for understanding why auxiliary tasks targeting specific structure types may improve overall performance. Table 4 provides the complete class listing with clinical context.

Table 4. ToothFairy3 Anatomical Region Classification

Category	Count	Label IDs	Structures	Volume %	Clinical Significance
Jawbones	2	1, 2	Lower jawbone (mandible), Upper jawbone (maxilla)	30–40%	Orthognathic surgery planning, bone volume assessment for implants, fracture evaluation
Sinuses & Airway	3	5, 6, 7	Left/right maxillary sinus, pharynx	5–15%	Sinus lift planning, airway assessment, proximity evaluation for molar implants

Individual Teeth	32	11–48	All permanent teeth (FDI 11–18, 21–28, 31–38, 41–48)	15–20%	Tooth identification, orthodontic planning, extraction assessment, root morphology
Tooth Pulps	32	111–148	Pulp cavities for all 32 teeth	< 1%	Root canal treatment planning, pulp vitality assessment, endodontic access design
Neurovascular Canals	5	3, 4, 103, 104, 105	Left/right inferior alveolar canal, left/right mandibular incisive canal, lingual canal	< 0.1%	Implant safety zones, nerve preservation during surgery, prevention of permanent sensory loss
Dental Restorations	3	8, 9, 10	Bridge, crown, implant	Variable	Treatment history documentation, artifact identification, prosthetic planning

Volume percentages are approximate and vary across patients depending on dentition status, anatomy, and pathology. Not all 77 classes are present in every scan — edentulous patients lack some tooth and pulp cavity classes, and not all patients present with wisdom teeth. Dental restoration classes (bridge, crown, implant) are included in the dataset annotations but were excluded from the challenge ranking metrics [43].

For disaggregated metric reporting, the 77 classes are grouped into four evaluation tiers based on volumetric proportion and segmentation difficulty.

Table 5. Evaluation Size Categories for Disaggregated Metric Reporting

Tier	Structures (# classes)	Metrics Reported	Rationale
Large (5 classes)	Mandible, maxilla, L/R sinuses, pharynx	DSC, HD95	Occupy 30–55% of volume; standard metrics sufficient
Medium (32 classes)	32 individual teeth	DSC, HD95	Moderate volume (15–20%); apex landmarks auxiliary task targets these
Small—Pulps (32 classes)	32 tooth pulp cavities	DSC, HD95	Tiny structures inside teeth; severe class imbalance; benefits from MTL
Small—Tubular (5 classes)	L/R IAC, L/R incisive canal, lingual canal	DSC, HD95, clDice	< 0.1% volume; tubular topology critical; centerline aux task targets these; clDice required

This four-tier scheme ensures that aggregate metrics across all 77 classes do not mask failures on clinically critical small structures, that clDice is reported specifically for the tubular neurovascular structures where topological connectivity matters, and that the contribution of each auxiliary task can be assessed against its target structures — apex landmarks for teeth, centerlines for canals.

3.3.5 Preprocessing Pipeline

All CBCT volumes from both ToothFairy3 and STS-3D-Tooth undergo identical preprocessing. First, volumes are resampled to isotropic 0.3×0.3×0.3 mm voxel spacing using trilinear interpolation for images and nearest-neighbour interpolation for label maps, matching the spacing used by U-Mamba2 [7].

Hounsfield Unit values are then clipped to the per-volume [0.5th, 99.5th] percentile range to suppress outlier intensities from metallic artifacts, then normalized to zero mean

and unit variance. Per-volume normalization is used rather than a global standard to accommodate the scanner-specific intensity distributions across the two datasets.

Non-zero foreground regions are cropped with a 10-voxel margin to strip empty background, which reduces memory load and keeps computation focused on anatomically relevant tissue.

During training, random 128×256×256 patches are extracted from the preprocessed volumes following U-Mamba2's patch configuration [7]. During inference, the full volume is processed using sliding-window inference with 50% overlap.

3.4. Automatic Auxiliary Label Generation

This section describes the computational pipeline for generating auxiliary task labels from existing primary segmentation masks. Both operations are applied only to the 106 labelled training scans from ToothFairy3, where 77-class ground-truth segmentation masks are available. No manual annotation is required beyond what already exists for the primary task.

The two auxiliary tasks target complementary subsets of the 77 classes. The tooth apex landmark task draws from all 32 individual tooth classes (FDI 11–48, label IDs 11–48), generating root tip coordinates for each tooth present in a given scan. Multi-rooted molars (FDI 16–18, 26–28, 36–38, 46–48) may produce 2–3 apex points per tooth. The total number of apex landmarks per scan varies by dentition status but typically ranges from 28 to 64 (approximately 28 present teeth × 1–2 roots on average, excluding missing or unerupted teeth). The nerve centerline task draws from all 5 neurovascular canal classes: the left and right inferior alveolar canals (labels 3, 4), the left and right mandibular incisive

canals (labels 103, 104), and the lingual canal (label 105). The expansion from ToothFairy2, which had only 2 inferior alveolar canals, to ToothFairy3's 5 canal classes increases the number of tubular structures available for centerline extraction and provides a richer auxiliary training signal for the centerline decoder head.

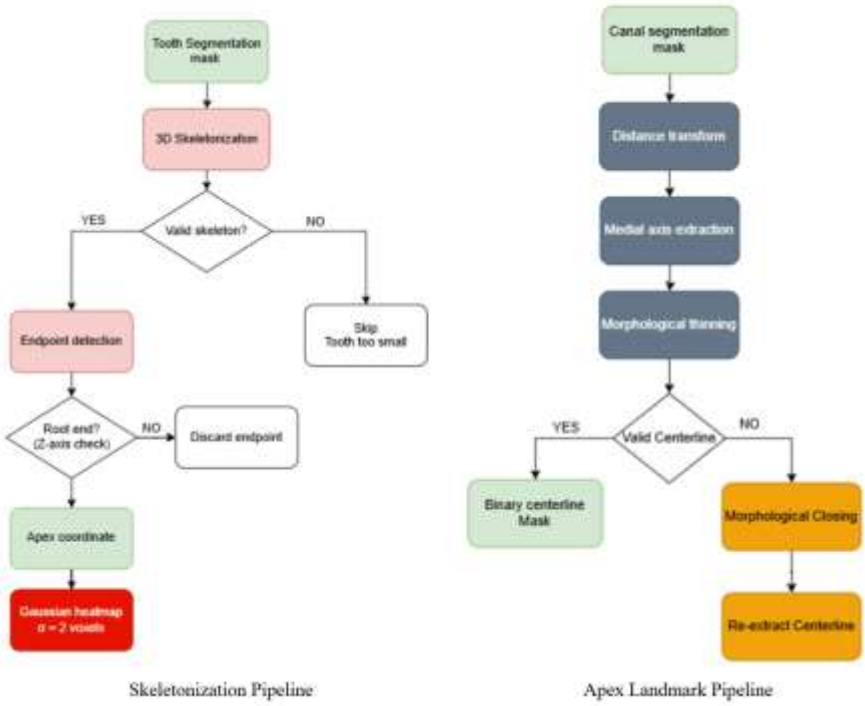

Figure 9. Automatic Auxiliary Label Generation Pipeline

3.4.1 Tooth Apex Landmark Extraction via Morphological Skeletonization

Tooth apex landmarks or the root tips are clinically critical reference points for endodontic procedures, requiring localization within 1–2 mm for safe treatment [16]. Landmark extraction operates on individual tooth segmentation masks through a five-step pipeline.

Table 6. Auxiliary Task Source Classes

Auxiliary Task	Source Classes from TF3	# Source Classes	Output Label Type
Tooth Apex Landmarks	32 individual tooth classes (FDI 11–48, labels 11–48)	32	3D Gaussian heatmap (σ=2 voxels)
Nerve Centerlines	L/R inferior alveolar canal (3,4), L/R incisive canal (103,104), lingual canal (105)	5	Binary centerline mask (1-voxel wide)

For each of the 32 permanent tooth classes in the segmentation mask, a binary volume is first isolated representing tooth versus background. The Lee-Kashyap-Chu 3D medial axis thinning algorithm [37] is then applied to each binary mask, iteratively removing surface voxels while preserving topological connectivity until the three-dimensional tooth structure is reduced to a one-voxel-wide skeleton along its medial axis. This is implemented via scikit-image's skeletonize_3d() function.

Skeleton endpoints are subsequently identified as voxels with exactly one 26-connected neighbor, with each tooth skeleton typically yielding two endpoints: one at the crown tip and one at the root apex. These endpoints are then classified by anatomical position: for maxillary teeth, the apex is the endpoint with the highest z-coordinate (closest to the skull base), while for mandibular teeth it is the one with the lowest z-coordinate (furthest from the occlusal plane). This decision rule exploits the consistent anatomical orientation of CBCT acquisitions. Multi-rooted teeth such as molars may produce multiple apex endpoints, all of which are retained.

Finally, a 3D Gaussian kernel ($\sigma = 2$ voxels, equivalent to 0.6 mm at 0.3 mm isotropic spacing) is placed at each detected apex to produce a continuous landmark

heatmap. The Gaussian representation is preferred over binary point labels because it yields smoother training gradients and naturally accommodates the 1–2 mm localization tolerance accepted in clinical practice.

3.4.2 Nerve Centerline Extraction via Distance Transform and Thinning

Nerve centerline extraction produces one-voxel-wide medial axes of tubular canal structures, capturing the topological connectivity that volumetric segmentation alone may miss. The pipeline is applied to the inferior alveolar canal, incisive nerve, and mental foramen canal masks through a four-step procedure.

A binary volume is first isolated for each neurovascular structure class from the segmentation mask. The Euclidean distance transform is then computed, assigning each foreground voxel its distance to the nearest background voxel. The resulting distance map carries peak values along the canal's medial axis and decreasing values toward its boundaries, effectively encoding the cross-sectional geometry of the structure.

Iterative morphological thinning is subsequently applied to reduce the canal mask to a one-voxel-wide centerline while preserving topological connectivity, implemented via scipy.ndimage for the distance transform and scikit-image's skeletonize_3d() for the thinning step [36], [37]. The output is a binary centerline mask in which medial axis voxels are labelled 1 and all others 0. This centerline serves as the ground-truth label for the nerve centerline auxiliary task head.

Table 7. Automatic Auxiliary Label Generation Performance Summary

Auxiliary Task	Input	Method	Output

Tooth Apex Landmarks	Tooth segmentation masks	3D skeletonization + endpoint filtering [37]	Gaussian heatmap (σ = 2 voxels)
Nerve Centerlines	Canal segmentation masks	Distance transform + thinning [36], [37]	Binary centerline mask

3.5. Model Architecture: Multi-Task U-Mamba2

This study employs U-Mamba2 [16] as the foundational segmentation architecture, extending it with two additional task-specific decoder heads to enable simultaneous prediction of primary anatomical segmentation, tooth apex landmark coordinates, and nerve centerline masks within a unified multi-task learning framework. U-Mamba2 integrates Mamba2 state space model blocks [30], [31] within a U-Net-inspired encoder-decoder topology [24], achieving linear computational complexity with respect to sequence length while capturing long-range spatial dependencies essential for relating distant anatomical structures in volumetric CBCT data. The architectural design implements the multi-task learning paradigm through a shared encoder that must learn feature representations benefiting all three prediction tasks, with task-specific decoder branches providing specialized processing for each output modality. This section describes the base U-Mamba2 architecture inherited from prior work, the multi-task extensions introducing auxiliary decoder heads, and the architectural parameters governing computational requirements and inference efficiency.

3.5.1 Base U-Mamba2 Architecture

The backbone architecture is U-Mamba2 [7], a hybrid CNN–state space model built within the nnU-Net framework. U-Mamba2 uses a seven-stage encoder-decoder in which

CNN-based residual blocks handle all stages except the bottleneck, where Mamba2 blocks [27] are placed for long-range dependency modelling. The architecture processes 128×256×256 input patches at 0.3 mm isotropic spacing.

Several design choices from U-Mamba2 are preserved in this study. Mamba2 blocks are positioned exclusively at the lowest-resolution bottleneck stage, where reduced spatial dimensions keep the sequence length manageable for the state space mechanism while the resulting global context propagates to all decoder stages via skip connections [7]. To address class imbalance, structures occupying less than 0.1% of voxels — incisive nerves, mental foramina, and the lingual foramen — receive elevated loss weights (10×) so that their gradients contribute proportionally to model updates [7]. Bilateral symmetry augmentation swaps labels for left-right anatomical pairs whenever horizontal flipping is applied, preventing the model from learning anatomically impossible configurations [7]. Finally, label smoothing is applied between anatomically related classes, left-right counterparts and adjacent teeth, to encode semantic similarity in the supervision signal [7].

3.5.2 Multi-Task Extension: Three Task-Specific Decoder Heads

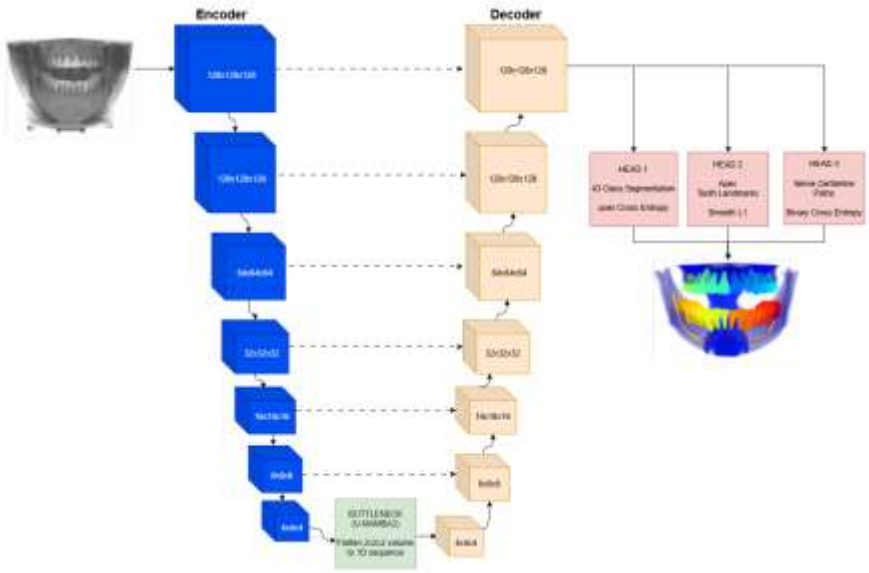

Figure 10. Proposed multi-task U-Mamba2 architecture extending the base U-Mamba2

The multi-task extension adds two lightweight decoder heads alongside the primary segmentation head, all sharing the common encoder and bottleneck features:

Head 1 — Primary Segmentation: A standard decoder producing a multi-class softmax probability map across all annotated anatomical classes. This head uses the full decoder with skip connections from all encoder stages.

Head 2 — Tooth Apex Landmark Prediction: A lightweight convolutional head (two 3×3×3 convolution layers followed by a 1×1×1 convolution with sigmoid activation) that produces a single-channel heatmap representing the probability of each voxel being a tooth apex. This head receives features from the shared decoder's final layer.

Head 3 — Nerve Centerline Prediction: A lightweight convolutional head with identical architecture to Head 2, producing a single-channel binary probability map representing the predicted nerve centerline. Sigmoid activation is used since this is a binary prediction task.

The lightweight design of Heads 2 and 3 (two convolutional layers each) adds minimal computational overhead—less than 2% additional parameters relative to the base U-Mamba2. Their primary function is to provide auxiliary gradient signals that flow back through the shared encoder during training, enriching the encoder's feature representations without substantially increasing model complexity or inference time.

3.6. Semi-Supervised Learning Framework: Mean Teacher

3.6.1 Mean Teacher Training Protocol and Multi-Task Consistency Regularization

The Mean Teacher framework [28] maintains two copies of the multi-task U-Mamba2 model: a Student network (θ_S) updated via gradient descent, and a Teacher network (θ_T) updated as an exponential moving average (EMA) of the Student's weights.

For each labeled CBCT volume, the Student produces three outputs — a multi-class segmentation prediction, a tooth apex landmark heatmap, and a nerve centerline probability map — each compared against its corresponding ground-truth label: the segmentation prediction against the expert-annotated mask, the apex prediction against the auto-generated Gaussian heatmap, and the centerline prediction against the auto-generated binary centerline mask. The supervised loss is the weighted sum of these three task-specific losses.

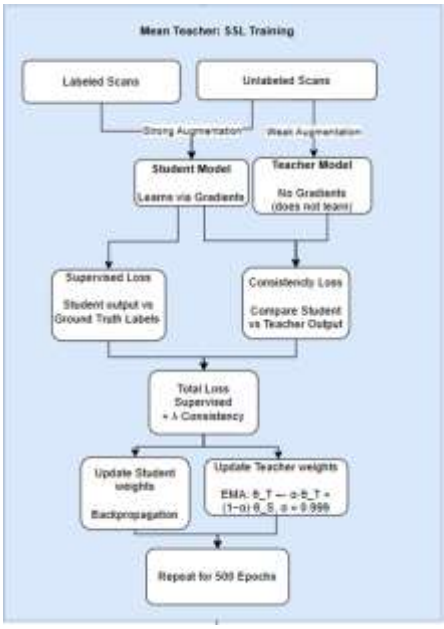

Figure 11. Mean Teacher Pipeline

For each unlabeled volume drawn from both ToothFairy3 and STS-3D-Tooth, two augmented versions are created: a weakly augmented version (random flip, small rotation) fed to the Teacher, and a strongly augmented version (elastic deformation, intensity jitter, random erasing) fed to the Student. Both models produce three outputs each across all task heads. The consistency loss enforces that the Student's predictions on the strongly augmented input match the Teacher's predictions on the weakly augmented input, computed separately per task head. The Teacher's outputs serve as pseudo-labels — they are not compared against any ground truth, but act as stable regression targets that the Student must learn to reproduce despite the stronger perturbation applied to its input.

After each Student gradient update, the Teacher's weights are updated via EMA:

$$\theta_T \leftarrow \alpha \cdot \theta_T + (1 - \alpha) \cdot \theta_S \qquad (1)$$

where α = 0.999. The Teacher never receives gradient updates directly. This smoothing mechanism produces a temporally averaged version of the Student whose predictions are more stable and reliable as pseudo-labels than those of the Student itself, whose weights change rapidly during training [28].

3.6.2 Task-Specific Loss Functions and Multi-Task Weighting

The total training loss combines supervised and consistent components across all three tasks.

$$L_{total} = L_{\sup ervised} + \lambda \cdot L_{consistency} \quad (2)$$

$$L_{\sup ervised} = L_{seg} + \lambda_{apex} \cdot L_{apex} + \lambda_{cl} \cdot L_{centerline} \quad (3)$$

$$L_{apex} = \frac{1}{N}\sum_{i}^{0}(\widehat{y_i} - y_i)^2 \quad (4)$$

$$L_{centerline} = L_{BCE} + \lambda_{topo} \cdot L_{soft-clDice} \quad (5)$$

The segmentation loss (L_seg) follows U-Mamba2's weighting scheme [7], combining Dice loss and cross-entropy with class-specific weights: Dice loss handles class imbalance by computing per-class overlap, while cross-entropy provides stable gradients for rare structures.

$$L_{CE} = -\frac{1}{N}\sum_{i}^{0}\sum_{c}^{0} w_c g_{i,c} \log(p_{i,c})$$

(6)

where w_c = class weight for class c, N = number of voxels

The apex landmark loss (L_apex) is mean squared error (MSE) between the predicted heatmap and the auto-generated Gaussian heatmap, which is appropriate given the continuous nature of the regression target. The centerline loss (L_centerline) combines binary cross-entropy with soft-clDice loss [36], where the soft-clDice component explicitly

penalizes fragmentation and encourages topologically connected centerline predictions. Consistency losses for each task head are computed as MSE between Student and Teacher predictions and applied only to unlabeled data.

The auiliary tasks are weighted at λ_apex = λ_cl = 0.5, keeping them subordinate to the primary segmentation objective so that auxiliary gradients enrich the shared encoder without dominating optimization. The consistency weight λ_cons ramps linearly from 0 to 1.0 over the first 30% of training epochs, following a warm-up schedule that allows the Teacher's predictions to stabilize before they are heavily relied upon as pseudo-labels [8].

3.6.3 Data Augmentation Strategy: Weak Versus Strong Perturbations

Data augmentation applies stochastic spatial and intensity transformations to training patches, improving robustness and reducing overfitting. The augmentation strategy distinguishes weak versus strong augmentations following FixMatch principles [46]: weak augmentations preserve anatomical structure with minimal distortion (used for teacher inputs providing stable pseudo-targets), while strong augmentations apply aggressive perturbations (used for student inputs forcing invariant representation learning). All augmentations preserve anatomical plausibility by maintaining spatial relationships, using smooth displacement fields, and exchanging bilateral structure labels during mirroring to prevent anatomically impossible configurations [16].

Table 8. Data Augmentation Strategy for Weak and Strong Perturbations

Augmentation	Weak (Teacher Input)	Strong (Student Input)
Random flip	Horizontal/vertical, p = 0.5	Horizontal/vertical, p = 0.5

Random rotation	±5°	±15°
Elastic deformation	None	$\alpha = 200$, $\sigma = 20$, $p = 0.3$
Intensity jitter	None	Brightness ±0.2, Contrast ±0.2
Gaussian noise	None	$\sigma = 0.1$, $p = 0.2$
Random erasing	None	10–20% of patch, $p = 0.3$
Label swap	Applied when flipping	Applied when flipping

3.7. Training Protocol and Hyperparameters

Both Configuration A (single-task baseline) and Configuration B (multi-task proposed) are trained with identical hyperparameters to ensure controlled comparison. Training is conducted on Google Colab using NVIDIA A100 (40 GB) GPU. Key hyperparameters follow U-Mamba2's published configuration [7] with modifications for the SSL protocol.

Table 9. Complete Hyperparameter Specification

Parameter	Value
Optimizer	AdamW (weight decay 1e-5)
Initial learning rate	1e-3 with cosine annealing to 1e-6
Batch size	2 (per GPU)
Patch size	128 × 256 × 256
Training epochs	500 (with early stopping, patience = 50)
EMA decay (α)	0.999
Consistency ramp-up	Linear, 0 → 1.0 over first 150 epochs
Auxiliary task weights	$\lambda_apex = 0.5$, $\lambda_cl = 0.5$

Inference	Sliding window, overlap 0.5, TTA (mirroring)
Target GPU	NVIDIA A100 40GB (training), T4 16GB (inference)
Mixed precision	FP16 (automatic mixed precision)

3.8. Evaluation Metrics and Analysis

3.8.1. Primary Metrics

Model performance is quantified through two standard medical image segmentation metrics universally accepted in the literature: Dice Similarity Coefficient measuring volumetric overlap between predictions and ground truth, and Hausdorff Distance 95th percentile measuring boundary localization accuracy [7], [16]. The Dice Similarity Coefficient for anatomical class c is computed where Pc represents the set of voxels predicted as class c, Gc represents the set of voxels labeled as class c in ground truth, |·| denotes set cardinality (voxel count), and |Pc ∩ Gc| counts true positive voxels correctly predicted as class c. DSC ranges from 0 (no overlap, complete prediction failure) to 1 (perfect agreement), with values ≥0.80 considered clinically acceptable for diagnostic and treatment planning applications based on established medical imaging standards [7], [16]. Mean DSC averaged across all 43 anatomical classes provides overall performance summary enabling direct comparison between configurations, while per-class DSC values enable structure-specific analysis identifying which anatomies benefit most from multi-task learning.

$$DSC_c = \frac{2 \cdot |P_c \cap G_c|}{|P_c| + |G_c|} \quad (7)$$

where Pc and Gc denote predicted and ground truth voxel sets for class c, with the numerator computing twice the intersection (true positives) and denominator summing

individual set cardinalities. The factor of 2 in the numerator ensures DSC=1 when prediction exactly matches ground truth.

Hausdorff Distance 95th percentile (HD95) measures the maximum boundary localization error while excluding the worst 5% outliers to improve robustness against small errors or isolated mispredictions [7]. For each anatomical class c, boundary voxels are extracted from both prediction and ground truth via morphological gradient (voxels adjacent to different classes), then for each predicted boundary point p, the distance to the nearest ground truth boundary point is computed via Euclidean norm, and vice versa for each ground truth boundary point g to predicted boundaries. The 95th percentile of this bidirectional distance distribution constitutes HD95, reported in millimeters. Lower HD95 values indicate better boundary precision; typical clinical thresholds are HD95 <15 mm considered acceptable and HD95 <10 mm considered good for dental CBCT applications [16]. HD95 provides complementary information to Dice coefficient because models may achieve high volumetric overlap (high Dice) while exhibiting poor boundary localization (high HD95), or vice versa.

$$HD_{95} = \max\left(d_{95}(P \rightarrow G), d_{95}(G \rightarrow P)\right) \quad (8)$$

Centerline Dice (clDice) evaluates topological connectivity of tubular structures. Both the prediction and ground truth are skeletonized to one-voxel-wide centerlines. clDice computes the harmonic mean of topology precision (fraction of predicted skeleton within the ground-truth volume) and topology sensitivity (fraction of ground-truth skeleton within the predicted volume) [36]. Clinical acceptance threshold: clDice ≥ 0.75 for neurovascular structures. This metric directly addresses panelist concerns that DSC and HD95 alone cannot capture whether nerve canal predictions are topologically intact—a fragmented

canal with acceptable DSC would score poorly on clDice, correctly reflecting its clinical inutility.

$$clDice = \frac{2 \cdot T_{prec} \cdot T_{sens}}{T_{prec} + T_{sens}} \quad (9)$$

3.8.2. Structure Specific Analysis

Results are disaggregated across three anatomical size categories to ensure that aggregate metrics do not mask failures on clinically critical small structures

Table 10. Anatomical Size Categories and Clinical Significance

Category	Structures	Volume Proportion	Clinical Significance
Large	Mandible, maxilla, maxillary sinuses	30–40% of scan volume	Surgical planning, bone assessment
Medium	Individual teeth (32 classes), pulp chambers	15–20% of scan volume	Tooth identification, endodontic planning
Small	Inferior alveolar canal, incisive nerve, mental foramina, lingual foramen	< 0.1% of scan volume	Implant safety, nerve preservation

All three metrics (DSC, HD95, clDice) are reported for each category. clDice is particularly important for the Small category, where topological connectivity is the primary clinical concern.

Table 11. Evaluation Metrics and Clinical Acceptance Thresholds

Metric	What It Measures	Threshold	Best For	Limitation

DSC	Volumetric overlap	≥ 0.80	All structures	Ignores topology, biased to large structures
HD95	Boundary distance (95th percentile)	< 15 mm	Boundary localization	Does not penalize connectivity gaps
clDice	Centerline overlap (topology)	≥ 0.75	Tubular neurovascular structures	Requires skeletonization step

3.9. Role of Domain Experts

Domain expert validation provides a clinical usability assessment that quantitative metrics alone cannot capture. Aggregate DSC, HD95, and clDice scores may indicate acceptable average performance while masking localized failures that would be unacceptable in practice — a nerve canal prediction that is volumetrically accurate but anatomically implausible in its trajectory, for instance, would pass quantitative thresholds while being clinically unusable. Expert review is designed to catch precisely these failure modes.

Experts serve three functions in this study. They perform qualitative grading by reviewing segmentation outputs on a subset of test scans and rating them on a 5-point Likert scale (1 = clinically unacceptable, 5 = clinically excellent) across anatomical accuracy, boundary precision, nerve canal continuity, and overall utility for treatment planning. They also identify error categories that metrics may not surface, including anatomically

implausible predictions, false connectivity between adjacent structures, and systematic regional biases. Finally, before training begins, experts review a random sample of ToothFairy3 ground-truth annotations to confirm label quality; this review is documented in the appendix alongside each expert's letter of participation.

Two domain experts are recruited to enable inter-rater reliability assessment via Cohen's kappa coefficient, which quantifies agreement between independent evaluations and guards against individual grading bias. Eligible experts are licensed dentists with documented experience in CBCT interpretation. Board-certified oral and maxillofacial radiologists are preferred, though general dentists with regular CBCT use for implant planning or endodontic assessment are acceptable. Local dentists in the Philippines meet these criteria provided they satisfy the CBCT experience requirement.

Dr. Eloisa Viloria is a Principal Dentist and oral specialist. Her clinical expertise is particularly relevant to this study's focus areas: delineating complex dental structures such as root apices for endodontic planning and tracing the location and course of the inferior alveolar nerve. Dr. Viloria participates under a formal consultancy agreement acknowledging the research purpose, data confidentiality requirements, and appropriate attribution in resulting publications. Compensation or co-authorship credit is offered to commensurate with the scope of her contribution.

3.10. Deployment Scope

This study is a proof-of-concept research investigation, and the distinction between proof-of-concept and clinical deployment is important for setting appropriate expectations. This study evaluates whether the proposed multi-task SSL framework produces measurable

segmentation improvements under controlled experimental conditions. It does not produce a system intended for direct clinical use on patients. The proof-of-concept scope means the following deliverables are produced: trained model weights for both Configuration A and Configuration B, quantitative evaluation results on the held-out ToothFairy3 test set across all metrics and anatomical tiers, domain expert qualitative grading of segmentation outputs, the automatic auxiliary label generation pipeline code, and computational feasibility metrics.

To demonstrate that the approach is computationally viable for eventual clinical translation, several feasibility metrics are reported. In terms of model complexity, the base U-Mamba2 model has approximately 30 to 50 million parameters depending on configuration, and the multi-task extension adds less than two percent additional parameters via two lightweight auxiliary decoder heads consisting of two 3x3x3 convolution layers each, with the total model file size remaining under 200 MB, small enough for distribution via standard file transfer and storage on clinical workstation hard drives. Regarding training requirements, the framework requires an NVIDIA A100 or equivalent GPU with mixed-precision support, with an estimated training time of three to five days for 500 epochs on the full 1,032-scan training pool, which is achievable on Google Colab Pro+ or comparable cloud computing platforms within the project timeline. On inference performance, U-Mamba2 achieves inference times of approximately six to ten seconds per full CBCT volume on an NVIDIA T4 GPU, which is representative of hardware available in modern dental radiology workstations and is well within clinical acceptability for same-visit workflows where the alternative is 25 to 45 minutes of manual segmentation.

Translating this proof-of-concept into a deployable clinical tool would require several additional steps that fall beyond the scope of this undergraduate thesis. At minimum, these would include regulatory clearance appropriate to the target jurisdiction, prospective validation on independent multi-site clinical cohorts with pre-registered study protocols, integration with PACS and dental practice management software for seamless workflow embedding, user acceptance testing with practicing dentists and radiologists to evaluate usability and trust, and ongoing post-deployment monitoring for distribution shift as scanner hardware, software updates, and patient populations evolve. These steps are documented as future work directions.

3.11. Timeline and Feasibility

Table 12. Complete 15-Week Study Timeline

Week	Phase	Deliverables
1–2	Dataset Acquisition & Setup	Download datasets; set up Colab; install dependencies; verify loading
2–3	Preprocessing & Splitting	Run preprocessing; create data splits; generate auxiliary labels
3–4	Baseline Implementation	Implement Config A; test training loop; fix bugs
4–5	Baseline Training	Train baseline model; validate; record metrics
5–6	Multi-Task Implementation	Add auxiliary heads; integrate losses; test pipeline

6–8	Multi-Task Training	Train Config B; validate; run ablation tests
8–9	Final Evaluation	Test both configs; compute metrics; generate visuals
9–10	Domain Expert Review	Collect expert ratings; compute agreement
10–12	Writing & Revision	Write results/discussion; finalize figures and references

Replicability is ensured through use of publicly available components and explicit specifications: ToothFairy3 dataset is publicly accessible under CC-BY-NC-SA license [38], [39], [40] requiring only user registration; U-Mamba2 base architecture is open-source with GitHub repository [16]; morphological algorithms are implemented via standard libraries (scikit-image skeletonization [22], scipy distance transform [33]) with version numbers specified; complete hyperparameter values are documented in Table 3.3; evaluation metrics follow standard definitions with computation protocols specified; and software environment specifications are provided (Python 3.10.12, PyTorch 2.0.1, library versions). Any researcher with Google Colab access and the ToothFairy3 dataset can replicate this methodology by following documented procedures, with expected results reproducible within statistical variability from random initialization and stochastic training dynamics.

3.12. Ethical Considerations

This study uses exclusively publicly available, de-identified datasets (ToothFairy3 and STS-3D-Tooth) released under research-permissible licenses for academic use. No patient-identifiable information is accessed or processed at any stage. Both datasets were collected and de-identified by their respective institutions with appropriate institutional

review board approval as documented in their publications [3], [9]. Domain expert participation involves professional evaluation of AI outputs, not human subjects research, and is documented through signed participation letters included in the appendix. Code, trained model weights, and the automatic label generation pipeline are planned for release under permissive open-source licensing to support reproducibility.

Chapter 5

RESULTS AND DISCUSSION

To be discussed in Capstone/Thesis 3…..

Chapter 6

CONCLUSION AND RECOMMENDATION

To be discussed in Capstone/Thesis 3..

APPENDICES

Appendix A

Letter of Commitment (Domain Expert Acceptance Dr. Eloisa Vilaria)

February 18, 2026

Eloisa Viloria

Subject: Letter of Commitment - Domain Expert

Dear Ms. Viloria,

This is to acknowledge our consent with you that you will serve as the Domain Expert to our undergraduate thesis project that shall be undertaken under Thesis 1, Thesis 2 and Thesis 3.By signing this letter, you signify your willingness to guide and support our group throughout the entire duration of the study by providing professional expertise relevant to the project domain.

As the Domain Expert, you will provide professional guidance on the medical interpretation of dental radiographs and the analysis of dental fractures. You will review our research approach and offer expert recommendations to help ensure that the proposed AI-based system remains clinically relevant, accurate, and valid.

Project Title: **Multi-Task Semi-Supervised Learning with Automatically Generated Auxiliary Labels for Dental CBCT Segmentation**

Scope of Responsibilities:

As the Domain Expert, your responsibilities may include, but are not limited to:

- Validating and reviewing the ground truth annotations of dental fractures in radiographic images.
- Providing expert feedback on the segmentation outputs generated by the proposed AI model.
- Assessing the clinical relevance and applicability of the system in real-world dental settings.
- Advising on domain-specific nuances during the data preparation, model evaluation, and testing phases of the project.

We sincerely look forward to your guidance and mentorship in helping us successfully complete this research and bring the project to fruition.

Thank you for your time and consideration.

Sincerely:

Alisandro N. Domingo **Jherrie Nathaniel C. Millena** **Ryan Osmon S. Patenia**

Conforme:

Eloisa Viloria
Domain Expert

Date:_____

Appendix B

Letter of Commitment (Domain Expert Acceptance - Dr. Ma. Ruth B. Alivert)

March 14, 2026

Dr. Ma. Ruth E. Añovert

Subject: Letter of Commitment - Domain Expert

Dear Dr. Añovert

This is to acknowledge our consent with you that you will serve as the Domain Expert to our undergraduate thesis project that shall be undertaken under Thesis 1, Thesis 2 and Thesis 3.By signing this letter, you signify your willingness to guide and support our group throughout the entire duration of the study by providing professional expertise relevant to the project domain.

As the Domain Expert, you will provide professional guidance on the medical interpretation of dental radiographs and the analysis of dental fractures. You will review our research approach and offer expert recommendations to help ensure that the proposed AI-based system remains clinically relevant, accurate, and valid.

Project Title: **Multi-Task Semi-Supervised Learning with Automatically Generated Auxiliary Labels for Dental CBCT Segmentation**

Scope of Responsibilities:

As the Domain Expert, your responsibilities may include, but are not limited to:

- Validating and reviewing the ground truth annotations of dental fractures in radiographic images.
- Providing expert feedback on the segmentation outputs generated by the proposed AI model.
- Assessing the clinical relevance and applicability of the system in real-world dental settings.
- Advising on domain-specific nuances during the data preparation, model evaluation, and testing phases of the project.

We sincerely look forward to your guidance and mentorship in helping us successfully complete this research and bring the project to fruition. We are ready to discuss your standard rates or agree upon a suitable honorarium that reflects the value of your time and mentorship across the Thesis 1-3 timeline

Thank you for your time and consideration.

Sincerely:

Alisandro N. Domingo | Jherrie Nathaniel C. Millena | Ryan Osmon S. Patenia

---

Conforme:

Dr. Ma. Ruth E. Añovert
Domain Expert

Date:

REFERENCES

[1] Khattak O, Ahmed N, Tsakos G. CBCT in orthodontic and maxillofacial imaging: A comprehensive review. J Oral Biol Craniofac Res. 2023;13(2):187–196.

[2] Almufarrij I, Alobaid A. Cone-beam computed tomography in dental practice: Clinical applications and limitations. Saudi Dent J. 2023;35(5):431–442.

[3] Cipriano M, Picioreanu I, Pintore F, et al. ToothFairy3: Multi-structure segmentation in CBCT volumes. Grand Challenge, MICCAI 2025. Available: https://toothfairy3.grand-challenge.org/

[4] Cipriano M, Giannoni L, Lombardi G, et al. Deep segmentation of the mandibular canal: A new dataset and a baseline model. In: Proc. MICCAI Workshop. Springer; 2022:460–469.

[5] Dot G, Chaurasia A, Dubois G, et al. DentalSegmentator: Robust open source deep learning-based CT and CBCT image segmentation. J Dent. 2024;147:105130.

[6] Zeoline S, Katsumata A, Hedderich DM, et al. Fully automatic AI segmentation of oral surgery-related anatomies in CBCT. Int J Oral Sci. 2024;16(1):1–12.

[7] Tan JW, Ooi YS, Chua KL. U-Mamba2: Scaling state space models for dental anatomy segmentation in CBCT. In: Proc. ToothFairy3 Challenge, MICCAI. 2025. arXiv:2509.12069.

[8] Tan JW, Ooi YS, Chua KL. U-Mamba2-SSL for semi-supervised tooth and pulp segmentation in CBCT. In: Proc. STSR Challenge, MICCAI. 2025.

[9] Wang Y, Zhang Y, Chen X, et al. A multi-modal dental dataset for semi-supervised deep learning image segmentation. Sci Data. 2025;12:117.

[10] Yang X, Song Z, King I, Xu Z. A survey on deep semi-supervised learning. IEEE Trans Knowl Data Eng. 2023;35(9):8934–8954.

[11] Jiao R, Zhang Y, Ding L, et al. Learning with limited annotations: A survey on deep semi-supervised learning for medical image segmentation. Comput Biol Med. 2024;169:107840.

[12] Vandenhende S, Georgoulis S, Van Gansbeke W, et al. Multi-task learning for dense prediction tasks: A survey. IEEE Trans Pattern Anal Mach Intell. 2022;44(7):3614–3633.

[13] Chen Y, Xu K, Wu J, et al. Multi-task fully convolutional network for tooth instance segmentation in dental CBCT. IEEE Trans Med Imaging. 2022;41(11):3118–3130.

[14] Wang Y, Chen X, Qian D, et al. Hierarchical multi-task learning for dental CBCT segmentation. In: Proc. STS Challenge, MICCAI. 2023.

[15] Duan H, Zhu Y, Xu J, et al. Dual-task deep learning for root canal segmentation with boundary refinement. Dentomaxillofac Radiol. 2023;52(4):20220389.

[16] Rizwan M, Alshahrani S, Alshahrani A. Prospective evaluation of CBCT versus intraoral radiography for periapical pathology detection. Clin Oral Investig. 2023;27(3):1245–1256.

[17] Kanneppady SK, Kanneppady SS, Lakshmi R. Retrospective analysis of inferior alveolar canal identification and implant complications using CBCT. Int J Implant Dent. 2022;8:34.

[18] Cipriano M, Pintore F, Ferrante E, et al. ToothFairy dataset. Università degli Studi di Modena e Reggio Emilia. 2024. Available: https://ditto.ing.unimore.it/toothfairy3/

[19] Park J, Kim H, Lee J. Inferior alveolar nerve canal segmentation on CBCT using U-Net with frequency-domain attention. Bioengineering. 2024;11(4):354.

[20] Kim S, Lee M, Park J. Improving accuracy for inferior alveolar nerve segmentation using multi-label nnU-Net with active learning. Sci Rep. 2025;15:91725.

[21] Hedderich DM, Hoebel KV, Giese D, et al. AI-based detection of metallic artefacts in CBCT for quality control. Eur Radiol. 2023;33(2):1234–1243.

[22] Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. Nat Methods. 2021;18(2):203–211.

[23] Isensee F, Wald T, Ulrich C, et al. nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation. In: Proc. MICCAI. Springer; 2024:479–490.

[24] Hatamizadeh A, Tang Y, Nath V, et al. UNETR: Transformers for 3D medical image segmentation. In: Proc. WACV. IEEE; 2022:574–584.

[25] Hatamizadeh A, Nath V, Tang Y, et al. Swin UNETR: Transformer-based semantic segmentation for brain tumors. In: Proc. BrainLes Workshop, MICCAI. Springer; 2022:272–284.

[26] Gu A, Dao T. Mamba: Linear-time sequence modeling with selective state spaces. In: Proc. ICML. 2024.

[27] Dao T, Gu A. Transformers are SSMs: Generalized models and efficient algorithms through structured state space duality. In: Proc. ICML. 2024.

[28] Chen X, Yuan Y, Zeng G, Wang J. Semi-supervised semantic segmentation with cross pseudo supervision. In: Proc. CVPR. IEEE; 2021:2613–2622.

[29] Li Y, Chen J, Xie X, Ma K, Zheng Y. Self-loop uncertainty: A novel pseudo-label for semi-supervised medical image segmentation. In: Proc. MICCAI. Springer; 2023:614–623.

[30] Lyu F, Ye M, Ma AJ, et al. Region-aware instructive learning for semi-supervised tooth segmentation. arXiv:2505.03538. 2025.

[31] Wang Y, Zhang Y, Chen X, et al. STS MICCAI 2023 Challenge: Grand challenge on 2D and 3D semi-supervised tooth segmentation. arXiv:2407.13246. 2024.

[32] Wang Y, Zhang Y, et al. MICCAI 2023 STS Challenge: A retrospective study of semi-supervised tooth segmentation methods. Pattern Recognition. 2025;161:111280.

[33] Wang Y, Zhang Y, et al. MICCAI STS 2024 Challenge: Semi-supervised instance-level tooth segmentation in panoramic X-ray and CBCT images. Med Image Anal. 2026. arXiv:2511.22911.

[34] Crawshaw M. Multi-task learning 1997–2024: Part II Regularization and optimization. Harvard Data Sci Rev. 2025;7(1).

[35] Li B, Czarnecki WM, Pascanu R. Implicit regularization of multi-task learning and finetuning in overparameterized neural networks. In: Proc. ICLR. 2024.

[36] Shit S, Paetzold JC, Sekuboyina A, et al. clDice—A novel topology-preserving loss function for tubular structure segmentation. In: Proc. CVPR. IEEE; 2021:16560–16569.

[37] Lee TC, Kashyap RL, Chu CN. Building skeleton models via 3-D medial surface axis thinning algorithms. CVGIP: Graphical Models and Image Processing. 1994;56(6):462–478.

[38] Bansal S, et al. A comprehensive survey of Mamba architectures for medical image analysis: Classification, segmentation, restoration and beyond. arXiv:2410.02362. 2024.

[39] Heidari M, Ghorbani Kolahi S, Karimijafarbigloo S, et al. Computation-efficient era: A comprehensive survey of state space models in medical image analysis. arXiv:2406.03430. 2024.

[40] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-Net: Convolutional Networks for Biomedical Image Segmentation. In Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI). Springer, Cham, 234–241. https://doi.org/10.1007/978-3-319-24574-4_28

[41] Anna Liakh. 2025. 3D Dental X-Rays (CBCT) vs 2D Dental X-Rays: What's the Difference & When You Need Each. IDR Radiology. Retrieved from https://idradiology.com/3d-dental-x-rays-cbct-vs-2d-dental-x-rays/

[42] The Computer Vision Foundation. 2025. CVPR 2025 Virtual Poster Session. Retrieved from https://cvpr.thecvf.com/virtual/2025/poster/35175

[43] Bolelli F, Marchesini K, van Nistelrooij N, et al. Segmenting maxillofacial structures in CBCT volume. In: Proc. IEEE/CVF CVPR. IEEE; 2025.

[44] Wang Y, Zhang Y, Chen X, et al. STS-Tooth: A multi-modal dental dataset for semi-supervised deep learning image segmentation [Data set]. Zenodo. 2024. https://doi.org/10.5281/zenodo.10597292

[45] Bolelli F, Lumetti L, Vinayahalingam S, et al. Segmenting the inferior alveolar canal in CBCTs volumes: The ToothFairy challenge. IEEE Trans Med Imaging. 2024.

[46] Lumetti L, Pipoli V, Bolelli F, Ficarra E, Grana C. Enhancing patch-based learning for the segmentation of the mandibular canal. IEEE Access. 2024.